Reflexive Reinforcement Learning: A Unified Operator-Theoretic Framework for Co-Evolving Prediction, Policy, and Environment

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Abstract

Standard reinforcement learning assumes that environment dynamics are independent of the learning system, treating predictions as passive signals. However, in many real-world settings, including financial markets, multi-agent coordination, and adaptive decision systems, predictions influence behavior and thereby alter the evolution of the environment.

We introduce *Reflexive Reinforcement Learning*, a framework for learning in environments with endogenous dynamics, where prediction, policy, and state transitions are jointly coupled. We formalize this interaction through a reflexive operator that captures the feedback loop between predictive signals and environment evolution. Under this formulation, learning corresponds to identifying stable configurations of a coupled dynamical system rather than optimizing reward under fixed dynamics.

We develop algorithmic approaches that explicitly account for prediction-dependent transitions, including gradient-based methods with endogenous correction and fixed-point consistency objectives. We further derive theoretical conditions characterizing convergence, instability, and oscillatory behavior, identifying reflexive gain as the key quantity governing system stability.

Empirical results demonstrate that classical reinforcement learning methods become unstable in environments with predictive feedback, while reflexive methods maintain stable and consistent behavior. These findings suggest that modeling endogenous dynamics is essential for learning systems deployed in environments influenced by their own predictions.

1 Introduction

Modern machine learning and reinforcement learning (RL) frameworks are built on a foundational assumption: predictions are passive and do not influence the evolution of the system being modeled. In standard RL, environment dynamics are treated as exogenous, with transitions governed by a fixed kernel $P(s_{t+1} \mid s_t, a_t)$, independent of the learning system itself.

However, this assumption is violated in a broad class of real-world systems. In financial markets, predictive signals influence trading behavior and thereby reshape price dynamics. In multi-agent coordination, shared forecasts alter collective strategies and equilibrium outcomes. In algorithmic decision systems, predictions directly affect the behavior of agents whose actions determine future states. In such environments, prediction is not merely descriptive—it is causally active.

This creates a fundamental mismatch between classical learning formulations and environments in which predictions are embedded within the system's causal structure. Existing approaches, including model-based RL and multi-agent reinforcement learning (MARL), capture interaction and planning but continue to treat prediction and environment dynamics as separable components.

To address this limitation, we introduce **Reflexive Reinforcement Learning (RRL)**, a framework in which prediction, policy, and environment evolution are jointly coupled. We model system dynamics through a reflexive operator:

$$s_{t+1} = T\big(s_t, \pi(\rho(s_t))\big), \tag{1}$$

where a predictive oracle ρ generates signals that influence policy decisions, which in turn affect environment transitions. This formulation introduces **endogeneity** into the learning process, as predictions produced by the model become causal inputs to the system dynamics.

Under this perspective, learning is no longer equivalent to optimizing reward under fixed dynamics. Instead, it becomes the problem of identifying **stable configurations of a coupled dynamical system**, characterized as fixed points of the reflexive operator. This shift introduces new challenges, including instability, oscillatory behavior, and non-standard gradient pathways arising from prediction-dependent transitions.

Within this framework, we derive algorithmic approaches that explicitly account for endogenous feedback. These include **Endogenous Policy Optimization (EPO)**, which extends policy gradient methods to incorporate feedback through predictive signals, and **Fixed-Point Reinforcement Learning (FPRL)**, which reformulates learning as equilibrium discovery via operator consistency.

Empirically, we demonstrate that classical reinforcement learning methods fail in reflexive environments, exhibiting instability and suboptimal convergence, while reflexive methods achieve stable and consistent behavior. These results suggest that reflexive learning is not merely an extension of RL, but a necessary framework for environments in which predictions actively shape outcomes.

Contributions. This paper introduces a new class of reinforcement learning problems in which environment dynamics depend on predictive signals generated by the learning system itself.

- **Endogenous MDP Framework.** We formalize learning in environments with prediction-dependent transitions, defining a new class of Markov decision processes with endogenous dynamics.
- **Reflexive Operator Formulation.** We introduce a unified operator $\Phi(s) = T(s, \pi(\rho(s)))$ that couples prediction, policy, and environment evolution.
- **Stability Theory for Reflexive Systems.** We derive conditions for convergence, instability, and oscillation, identifying reflexive gain as the key quantity governing system behavior.
- **Reflexive Learning Algorithms.** We propose methods that account for prediction-dependent dynamics through gradient correction and fixed-point consistency.
- **Empirical Validation under Endogenous Feedback.** We demonstrate that classical RL methods become unstable when predictions influence dynamics, while reflexive methods maintain stability.

2 Problem Redefinition: From Passive Prediction to Reflexive Learning

2.1 Limitations of the Classical Formulation

Classical machine learning and reinforcement learning are built on the assumption that predictions do not influence the underlying system dynamics. In supervised learning, outcomes are modeled as

$$Y \sim P(Y \mid X), \tag{2}$$

where the predictive model estimates a conditional distribution independent of its own deployment.

Similarly, reinforcement learning assumes that environment dynamics are governed by a transition kernel

$$s_{t+1} \sim P(s_{t+1} \mid s_t, a_t), \tag{3}$$

where actions $a_t \sim \pi(a_t \mid s_t)$ are selected by the agent, while the transition function P remains independent of predictive signals produced by the learning system [14, 10].

This separation extends to multi-agent settings, where interaction is modeled through policies but not through predictive feedback affecting environment dynamics [8, 9].

These formulations implicitly assume that prediction is *passive* and does not alter the causal structure of the system. However, this assumption is violated in environments where predictive signals are disclosed and acted upon, leading to endogenous system evolution.

2.2 Reflexive Learning Systems

We define a *reflexive learning system* as a dynamical system in which prediction, policy, and environment evolution are jointly coupled through an endogenous feedback loop.

Let $\mathcal{S}$ denote the state space. At time t, the system evolves according to:

$$s_{t+1} = \mathcal{T}\big(s_t, \pi(\rho(s_t))\big), \tag{4}$$

where:

- $\rho : \mathcal{S} \to \mathbb{R}$ is a predictive oracle mapping states to signals,
- $\pi : \mathbb{R} \to \mathcal{A}$ is a policy conditioned on predictive signals,
- $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the environment transition operator.

Unlike classical formulations, the predictive signal $\rho(s_t)$ directly influences the action selection process and thereby affects the next state. Consequently, the predictive model becomes part of the system's causal mechanism.

2.3 Endogenous Prediction Loop

The defining feature of reflexive systems is the presence of an *endogenous prediction loop*:

$$s_t \;\to\; \rho(s_t) \;\to\; \pi(\rho(s_t)) \;\to\; s_{t+1}. \tag{5}$$

This feedback loop contrasts with classical learning systems, where predictions are external to the environment dynamics. In reflexive systems, prediction and evolution are inseparable.

2.4 Coupled Operator System

To formalize this interaction, we define a *coupled operator system*.

Definition 1 (Reflexive Operator)**.** Let ρ_θ and π_ϕ be parameterized functions. The reflexive operator is defined as

$$\Phi_{\theta,\phi}(s) = \mathcal{T}\big(s, \pi_\phi(\rho_\theta(s))\big). \tag{6}$$

This formulation transforms the learning problem into the analysis of an operator acting on the state space, connecting to classical fixed-point theory [1, 16].

2.5 Learning as Fixed-Point Discovery

In reflexive systems, stable behavior corresponds to states that are invariant under the reflexive operator.

Definition 2 (Reflexive Fixed Point)**.** A state $s^* \in \mathcal{S}$ is a reflexive equilibrium if

$$s^* = \Phi_{\theta,\phi}(s^*). \tag{7}$$

This notion generalizes equilibrium concepts in dynamical systems and game theory, where stable configurations correspond to fixed points of underlying mappings.

2.6 Approximate Equilibrium Regime

Exact fixed points may not exist or may be computationally intractable. We therefore define an approximate notion:

Definition 3 (Approximate Reflexive Equilibrium)**.** A state s is an ϵ-equilibrium if

$$\|\Phi_{\theta,\phi}(s) - s\| < \epsilon. \tag{8}$$

This approximation aligns with practical learning settings where convergence is achieved up to a tolerance threshold.

2.7 Implications for Learning

This redefinition introduces several fundamental departures from classical learning:

- **Non-separability:** Prediction, policy, and dynamics are coupled and cannot be optimized independently.
- **Endogenous dynamics:** Transition dynamics depend on predictive signals, violating classical Markov assumptions.
- **Equilibrium-centric learning:** Learning corresponds to identifying fixed points rather than maximizing reward alone.
- **Non-standard optimization:** Gradients must account for feedback through predictive and environmental pathways.

These properties define a new class of learning systems that extend beyond classical reinforcement learning and multi-agent frameworks, motivating the theoretical and algorithmic developments that follow.

3 Theoretical Framework: Existence, Instability, and Convergence

The reflexive formulation introduced in the previous section transforms learning into the study of a coupled operator $\Phi_{\theta,\phi}$ acting on the state space. This shift raises fundamental questions: *Do stable configurations exist? Under what conditions are they stable? And how does learning behave in their presence or absence?*

In this section, we develop a theoretical foundation addressing these questions through fixed-point analysis, stability theory, and approximate convergence regimes.

3.1 Existence of Reflexive Equilibria

We begin by examining whether reflexive equilibria exist. Such equilibria correspond to fixed points of the operator:

$$s^* = \Phi_{\theta,\phi}(s^*). \tag{9}$$

Existence depends on regularity properties of the operator, particularly continuity and compactness of the state space.

Theorem 1 (Reflexive Equilibrium Existence)**.** *Let $\mathcal{S} \subset \mathbb{R}^n$ be compact and convex. If the operator $\Phi_{\theta,\phi} : \mathcal{S} \to \mathcal{S}$ is continuous, then there exists at least one fixed point $s^* \in \mathcal{S}$ such that*

$$s^* = \Phi_{\theta,\phi}(s^*). \tag{10}$$

3.2 Convergence Under Controlled Reflexive Gain

We establish a sufficient condition under which reflexive learning dynamics remain stable despite prediction-dependent transitions.

Theorem 2 (Convergence under Controlled Reflexive Gain)**.** *Let $\Phi_{\theta,\phi}(s) = T(s, \pi_\phi(\rho_\theta(s)))$ be the reflexive operator. Assume:*

- *T is Lipschitz continuous in (s, a) with constants L_s, L_a,*
- *π_ϕ is Lipschitz in its input with constant L_π,*
- *ρ_θ is Lipschitz in s with constant L_ρ.*

Define the reflexive gain:

$$G := L_a L_\pi L_\rho. \tag{11}$$

If the controlled gain condition holds:

$$L_s + G < 1, \tag{12}$$

then:

1. *The operator $\Phi_{\theta,\phi}$ is a contraction,*
2. *There exists a unique fixed point s^*,*
3. *The iteration $s_{t+1} = \Phi_{\theta,\phi}(s_t)$ converges globally to s^*,*
4. *The convergence rate is geometric with factor $L_s + G$.*

Proof. From Lipschitz assumptions:

$$|\Phi(s) - \Phi(s')| = |T(s, \pi(\rho(s))) - T(s', \pi(\rho(s')))| \qquad \leq L_s|s - s'| + L_a|\pi(\rho(s)) - \pi(\rho(s'))|. \tag{13}$$

Using Lipschitz continuity of π and ρ:

$$|\pi(\rho(s)) - \pi(\rho(s'))| \leq L_\pi L_\rho |s - s'|. \tag{14}$$

Thus:

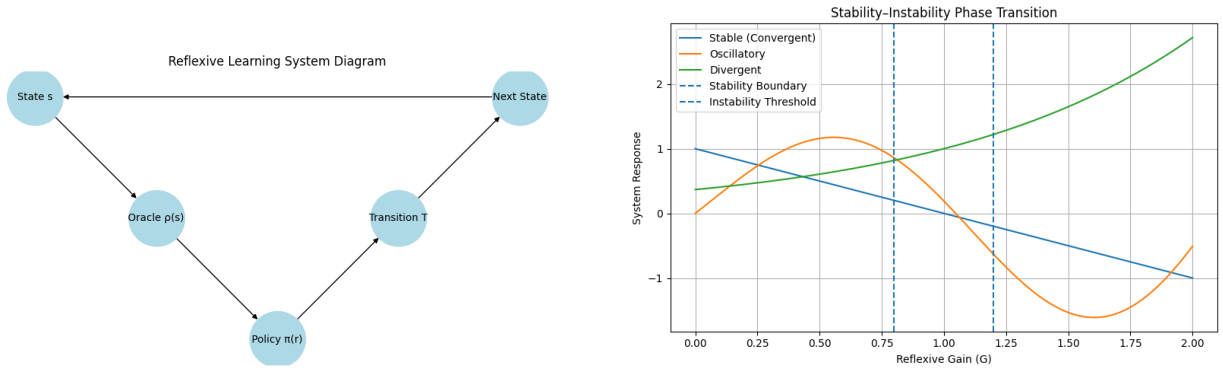
$$|\Phi(s) - \Phi(s')| \leq (L_s + L_a L_\pi L_\rho)|s - s'|. \tag{15}$$

If $L_s + G < 1$, then Φ is a contraction. The result follows from Banach's fixed-point theorem. □

Interpretation. This result shows that stability in reflexive systems depends not only on environment dynamics (L_s), but also on the strength of the prediction–policy–action feedback loop (G). Even if the environment itself is stable, excessive predictive sensitivity can break convergence.

Key implication. Stability can be enforced by explicitly controlling reflexive gain G, providing a principled foundation for regularization strategies in reflexive learning.

Interpretation. This follows from classical fixed-point results such as Schauder's theorem [12]. Intuitively, if the operator is continuous and the system remains within a bounded region, then a self-consistent configuration must exist. However, this result guarantees only existence, not uniqueness or stability.



(a) Reflexive system architecture

(b) Stability–instability phase transition

Figure 1: Reflexive learning systems exhibit a closed-loop interaction between prediction, policy, and environment dynamics (left), governed by a stability–instability phase transition controlled by reflexive gain (right). Increasing predictive influence shifts the system from stable convergence to oscillatory and divergent regimes.

3.3 Instability and Oscillation under Excessive Reflexive Gain

We now characterize conditions under which reflexive systems become unstable due to excessive predictive feedback.

Theorem 3 (Reflexive Instability Threshold)**.** *Let s^* be a fixed point of the reflexive operator $\Phi_{\theta,\phi}$, and assume $\Phi_{\theta,\phi}$ is differentiable at s^*.*

Let the Jacobian be:

$$D\Phi(s^*) = J_s + J_a J_\pi J_\rho, \tag{16}$$

where:

$$J_s = \frac{\partial T}{\partial s}, \quad J_a = \frac{\partial T}{\partial a}, \quad J_\pi = \frac{\partial \pi_\phi}{\partial r}, \quad J_\rho = \frac{\partial \rho_\theta}{\partial s}.$$

Define the reflexive gain:

$$G := |J_a J_\pi J_\rho|. \tag{17}$$

If:

$$\rho(D\Phi(s^*)) > 1, \tag{18}$$

then the fixed point s^* is unstable.

Moreover, a sufficient condition for instability is:

$$|J_s| + G > 1. \tag{19}$$

Proof. Local dynamics near s^* are governed by:

$$e_{t+1} = D\Phi(s^*)e_t. \tag{20}$$

If $\rho(D\Phi(s^*)) > 1$, perturbations grow exponentially, implying instability.

Using submultiplicativity:

$$|D\Phi| \leq |J_s| + |J_a J_\pi J_\rho| = |J_s| + G. \tag{21}$$

Thus, if $|J_s| + G > 1$, the operator norm exceeds unity, which implies $\rho(D\Phi) \geq 1$, leading to instability. □

Interpretation. This result shows that instability arises when predictive feedback amplification exceeds the intrinsic damping of the environment dynamics.

Key insight. Even if the environment is stable ($|J_s| < 1$), sufficiently large reflexive gain G can push the system into instability.

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Theorem 4 (Oscillatory Regime via Complex Eigenvalues)**.** *Let $\lambda = re^{i\theta}$ be an eigenvalue of $D\Phi(s^*)$.*

If:

$$r \approx 1 \quad \textit{and} \quad \theta \neq 0, \tag{22}$$

then the system exhibits oscillatory dynamics near s^*.

If additionally:

$$r > 1, \tag{23}$$

the oscillations grow in amplitude, leading to divergent cyclic behavior.

Proof. The linearized system evolves as:

$$e_t = \lambda^t e_0. \tag{24}$$

For complex $\lambda = re^{i\theta}$:

$$\lambda^t = r^t e^{it\theta}. \tag{25}$$

The magnitude evolves as r^t and the phase rotates by θ, producing oscillations. If $r > 1$, oscillations grow over time. □

Interpretation. Oscillations emerge from phase interactions introduced by predictive feedback, which can induce delayed or overcompensating responses in the system.

Connection to reflexivity. The term $J_a J_\pi J_\rho$ introduces coupling between prediction and action that can generate phase shifts, leading to cyclic dynamics.

3.4 Instability and Oscillatory Regimes

Even when fixed points exist, reflexive systems may exhibit instability or persistent oscillatory behavior. This arises when predictive feedback induces opposing responses that prevent convergence.

Rather than the absence of fixed points per se, a more common phenomenon is the presence of *non-contractive* or cyclic dynamics, where iterative application of $\Phi_{\theta,\phi}$ does not converge.

Observation. If the operator $\Phi_{\theta,\phi}$ is not a contraction and exhibits anti-monotonic or cyclic responses, then iterative updates of the form

$$s_{t+1} = \Phi_{\theta,\phi}(s_t) \tag{26}$$

may fail to converge, leading to oscillatory or chaotic trajectories.

Interpretation. For example, predictive signals that trigger overcompensating actions can systematically invalidate future predictions, producing feedback loops that oscillate indefinitely. Such behavior is closely related to cyclic dynamics studied in game-theoretic learning and non-contractive mappings [2, 5].

3.5 Local Stability and Sensitivity

Even when a fixed point exists, it may not be stable under perturbations. Let $D\Phi(s^*)$ denote the Jacobian of the operator at a fixed point.

Proposition 1 (Local Stability Criterion)**.** *A fixed point s^* is locally stable if the spectral radius of $D\Phi(s^*)$ satisfies*

$$\rho\big(D\Phi(s^*)\big) < 1. \tag{27}$$

Interpretation. This condition mirrors classical results from dynamical systems. If perturbations are amplified through the feedback loop (i.e., eigenvalues greater than one in magnitude), the system diverges from equilibrium. In reflexive systems, such amplification often arises when predictive signals exert excessive influence on agent behavior.

3.6 Approximate Convergence Regime

Given that exact convergence may be difficult to achieve, a more practical objective is approximate consistency:

$$\|\Phi_{\theta,\phi}(s) - s\| < \epsilon. \tag{28}$$

Theorem 5 (Approximate Convergence Under Contraction)**.** *Assume that $\Phi_{\theta,\phi}$ is Lipschitz continuous with constant $L < 1$. Then the iterative process*

$$s_{t+1} = \Phi_{\theta,\phi}(s_t) \tag{29}$$

converges to a unique fixed point s^*, with error bounded by

$$\|s_t - s^*\| \leq L^t \|s_0 - s^*\|. \tag{30}$$

Interpretation. This is a direct consequence of the Banach fixed-point theorem [1]. While strict contraction is uncommon in reflexive systems, this result characterizes a regime in which learning is stable and predictable. In practice, approximate convergence may still arise under regularization or stochasticity.

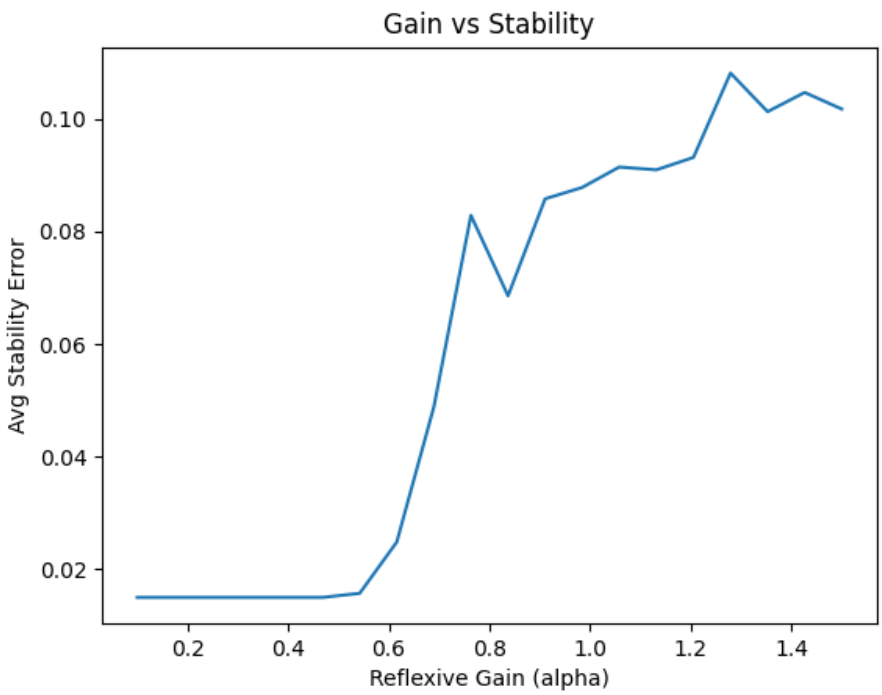


Figure 2: Effect of reflexive gain on stability. Increasing predictive influence leads to instability, consistent with the amplification behavior characterized by the Jacobian spectral radius.

3.7 Learning Under Parameter Updates

So far, we have treated $\Phi_{\theta,\phi}$ as fixed. In practice, both θ (oracle parameters) and ϕ (policy parameters) evolve during learning, causing the operator itself to change over time.

Proposition 2 (Two-Timescale Learning Dynamics)**.** *Suppose parameter updates (θ_t, ϕ_t) evolve on a slower timescale than state updates, and for each fixed parameter setting the operator $\Phi_{\theta,\phi}$ satisfies local contraction conditions. Then, under standard stochastic approximation assumptions, the system tracks a moving equilibrium and converges to a neighborhood of a reflexive fixed point [3].*

Interpretation. This aligns with classical two-timescale analysis: if the system stabilizes faster than parameters evolve, learning can adapt to shifting equilibria. If parameter updates are too aggressive, however, the system may fail to stabilize.

3.8 Summary of Theoretical Insights

The analysis above highlights several key properties of reflexive learning systems:

- **Existence is conditional:** Fixed points are guaranteed under continuity and compactness assumptions.
- **Instability is prevalent:** Non-contractive feedback can lead to oscillatory or divergent behavior.
- **Stability is sensitivity-dependent:** Small perturbations may be amplified through predictive influence.
- **Approximation is essential:** Practical learning often operates in approximate equilibrium regimes.
- **Learning is dynamic:** The operator evolves with parameter updates, complicating convergence analysis.

These results demonstrate that reflexive learning systems exhibit fundamentally different behavior from classical reinforcement learning. Understanding such systems requires integrating fixed-point theory, dynamical systems analysis, and adaptive optimization—motivating the algorithmic approaches developed in the following section.

4 Operator Formulation of Reflexive Learning Systems

The previous sections introduced the reflexive operator as a mapping over the state space. While this formulation is useful conceptually, real-world learning systems are inherently stochastic, multi-agent, and distributional. To capture these properties, we extend the reflexive operator into a distributional, operator-theoretic framework.

4.1 From State Dynamics to Distributional Operators

In practice, system states are not deterministic but are described by probability measures $\mu \in \mathcal{P}(\mathcal{S})$. The evolution of the system must therefore be defined at the level of distributions.

We define the distributional reflexive operator:

$$\Phi_{\theta,\phi}(\mu) = \mathrm{Law}\big(s_{t+1} \mid s_t \sim \mu, ; a_t \sim \pi_\phi(\rho_\theta(s_t))\big), \tag{31}$$

which maps a distribution over states to a new distribution after predictive signaling and policy response.

Interpretation. This formulation parallels distributional perspectives in reinforcement learning, but differs in that the transition dynamics depend explicitly on predictive signals. Instead of propagating value functions under fixed dynamics, the operator propagates state distributions under prediction-dependent evolution.

4.2 Relation to Classical Bellman Operators

The classical Bellman operator is given by:

$$\mathcal{B}V(s) = \mathbb{E}_{a\sim\pi,,s'\sim P}\big[r(s,a) + \gamma V(s')\big], \tag{32}$$

where transitions $P(s'|s,a)$ are assumed independent of the learning system.

In contrast, the reflexive formulation modifies the transition mechanism:

$$s_{t+1} \sim T\big(s_t, \pi(\rho(s_t))\big). \tag{33}$$

Key difference. The Bellman operator evaluates policies under fixed dynamics, whereas the reflexive operator captures settings in which the dynamics themselves depend on predictive signals. This introduces a feedback loop between learning and environment evolution.

4.3 Multi-Agent Reflexive Operators

Many reflexive systems involve multiple interacting agents. Let $\pi = (\pi_1, \ldots, \pi_N)$ denote agent policies. The operator generalizes to:

$$\Phi(\mu; \pi, \rho) = \mathrm{Law}\Big(s_{t+1} \mid s_t \sim \mu, ; a_i \sim \pi_i(\rho(s_t))\ \forall i\Big). \tag{34}$$

Interpretation. This connects to Markov games and multi-agent reinforcement learning [8, 9]. The distinguishing feature here is that agents condition on a shared predictive signal that depends on the state, introducing an additional layer of coupling not present in standard formulations.

4.4 Stochastic Reflexive Operators

To incorporate stochasticity in both prediction and decision-making, we express the operator in integral form:

$$\Phi(\mu) = \int T(s,a); d\pi(a \mid r); d\rho(r \mid s); d\mu(s). \tag{35}$$

Interpretation. This representation makes explicit the layered sources of randomness:

- uncertainty over states (μ),
- uncertainty in predictions (ρ),
- stochasticity in policy (π).

This aligns with measure-theoretic formulations of stochastic control [10], with the added feature that predictions mediate the transition process.

4.5 Reflexive Bellman Operator

To connect dynamics with value-based optimization, we define a reflexive Bellman operator:

$$\mathcal{B}^{\Phi}V(s) = \mathbb{E}\Big[r\big(s, \pi(\rho(s))\big) + \gamma V\big(\Phi(s)\big)\Big]. \tag{36}$$

Interpretation. This operator extends the classical Bellman formulation by incorporating prediction-dependent transitions. The resulting recursion couples value estimation with the evolution of the system itself.

4.6 Fixed Points in Distribution Space

In the distributional setting, equilibrium corresponds to a fixed point:

$$\mu^* = \Phi(\mu^*). \tag{37}$$

Interpretation. This generalizes equilibrium concepts from state space to distribution space, aligning with mean-field and population-level dynamics. Such equilibria describe invariant distributions under reflexive evolution [5].

4.7 Operator Geometry and Stability

Viewing Φ as an operator allows analysis of its geometric properties:

- **Contraction:** implies unique stable equilibria.
- **Non-contraction:** allows multiple equilibria or oscillatory behavior.
- **Expansiveness:** leads to instability.

These properties can be studied under metrics such as Wasserstein or total variation distance [18].

Key insight. Predictive feedback increases the sensitivity of the operator, making contraction less likely and instability more common than in classical reinforcement learning settings.

4.8 Implications

The operator formulation reveals several structural properties of reflexive learning systems:

- Learning is governed by a coupled dynamical system rather than a static optimization problem.
  “‘
- Predictive signals directly influence the evolution of the system, rather than serving as auxiliary inputs.
- Classical reinforcement learning operators emerge as special cases when predictive feedback is absent.
- Multi-agent interactions arise naturally within the operator through shared dependence on predictions.
  “‘

This perspective positions reflexive learning as an extension of reinforcement learning that explicitly accounts for prediction-dependent dynamics, motivating the algorithmic approaches developed in the next section.

5 Algorithms for Reflexive Learning

The operator formulation introduced earlier changes the structure of learning. In classical reinforcement learning, optimization assumes that environment dynamics are independent of the learning system. This allows gradients to be computed solely through the policy [14, 13].

In reflexive systems, this assumption no longer holds. Predictive signals influence agent behavior, and agent behavior in turn alters the environment dynamics. As a result, the transition operator depends on model parameters, introducing additional gradient pathways.

This section develops algorithmic approaches that explicitly account for this coupling.

5.1 Endogenous Gradient Policy (EGP)

In reflexive systems, actions are conditioned on predictive signals $r_t = \rho_\theta(s_t)$, and transitions depend on these signals:

$$s_{t+1} = T\big(s_t, \pi_\phi(\rho_\theta(s_t))\big). \tag{38}$$

Classical policy gradients assume exogenous transitions:

$$\nabla_\phi J = \mathbb{E}\left[\nabla_\phi \log \pi_\phi(a_t \mid s_t) \cdot A_t\right], \tag{39}$$

where A_t denotes an advantage estimate [15].

In reflexive settings, gradients must account for additional dependencies. Applying the chain rule yields:

$$\nabla_\phi J = \mathbb{E}\left[\nabla_\phi \log \pi_\phi(a_t \mid r_t) \cdot A_t\right] + \mathbb{E}\left[\frac{\partial A_t}{\partial r_t} \cdot \frac{\partial r_t}{\partial \phi}\right], \tag{40}$$

and for oracle parameters:

$$\nabla_\theta J = \mathbb{E}\left[\frac{\partial A_t}{\partial r_t} \cdot \frac{\partial r_t}{\partial \theta}\right] + \mathbb{E}\left[\frac{\partial A_t}{\partial s_{t+1}} \cdot \frac{\partial s_{t+1}}{\partial r_t} \cdot \frac{\partial r_t}{\partial \theta}\right]. \tag{41}$$

These terms capture:

- **Prediction sensitivity:** dependence of signals on parameters,
- **Environment sensitivity:** dependence of transitions on predictions.

We refer to this formulation as **Endogenous Gradient Policy (EGP)**. It extends policy optimization to settings where environment dynamics depend on predictive signals.

Update Rule.

$$\phi \leftarrow \phi + \alpha_\phi \nabla_\phi J, \;\; \theta \qquad\qquad \leftarrow \theta + \alpha_\theta \nabla_\theta J. \tag{42}$$

Interpretation. EGP does not introduce a new gradient estimator in isolation; rather, it extends policy optimization to endogenous environments. Ignoring these additional pathways can lead to instability or biased learning.

5.2 Fixed-Point Reinforcement Learning (FPRL)

The reflexive formulation shifts learning from reward maximization under fixed dynamics to identifying stable configurations of a coupled system:

$$\Phi_{\theta,\phi}(s) = T\big(s, \pi_\phi(\rho_\theta(s))\big). \tag{43}$$

A stable configuration corresponds to a fixed point:

$$s^* = \Phi_{\theta,\phi}(s^*). \tag{44}$$

We define a consistency objective:

$$\mathcal{L} * FP = |s - \Phi * \theta, \phi(s)|^2. \tag{45}$$

The optimization problem becomes:

$$\min_{\theta,\phi} \; \mathbb{E}\left[\mathcal{L}_{FP}\right]. \tag{46}$$

Interpretation. FPRL reframes learning as **equilibrium discovery**. Rather than optimizing behavior under fixed transitions, it seeks states that are invariant under prediction-dependent dynamics.

Connection to Classical RL. In standard RL, fixed points arise in value functions (e.g., Bellman equations). Here, fixed points are defined over *state evolution itself*, which depends on learned predictions.

Practical Role. FPRL is useful in systems where stability is critical. It can also be combined with reward-based objectives:

$$\mathcal{L} = -J + \lambda \mathcal{L}_{FP}. \tag{47}$$

5.3 Extensions of Reflexive Learning

The reflexive framework admits extensions that regulate predictive influence and system stability.

Information-Constrained Reflexivity. Constraining the mutual information between states and predictions limits excessive feedback, introducing a trade-off between informativeness and stability [17].

Adaptive Prediction Disclosure. A learned disclosure policy can control the strength of predictive signals, allowing the system to balance performance and stability dynamically.

Remark. These mechanisms are not standalone algorithms, but extensions that can be incorporated into EGP or FPRL.

5.4 Summary of Algorithmic Contributions

The proposed methods differ from classical reinforcement learning in two key ways:

- **Endogenous Gradient Policy (EGP):** accounts for gradient pathways induced by prediction-dependent dynamics.
- **Fixed-Point Reinforcement Learning (FPRL):** reformulates learning as equilibrium discovery under reflexive dynamics.

Together, these approaches provide a framework for learning in environments where prediction and system evolution are tightly coupled.

6 Algorithmic Implementation

6.1 Endogenous Gradient Policy (EGP)

Algorithm 1 Endogenous Gradient Policy (EGP)

0: Initialize oracle parameters θ, policy parameters ϕ
0: **for** each iteration **do**
0: Sample trajectories $\tau = (s_t)$
“‘
0: **for** each timestep t **do**
0: Compute predictive signal: $r_t \leftarrow \rho_\theta(s_t)$
0: Sample action: $a_t \sim \pi_\phi(a \mid r_t)$
0: Transition: $s_{t+1} \sim T(s_t, a_t)$
0: **end for**
0: Estimate advantages A_t
0: Compute policy gradient:

$$g_\phi \leftarrow \mathbb{E}\left[\nabla_\phi \log \pi_\phi(a_t \mid r_t) \cdot A_t\right]$$

0: Compute endogenous correction:

$$g_{\text{endo}} \leftarrow \mathbb{E}\left[\frac{\partial A_t}{\partial r_t} \cdot \frac{\partial r_t}{\partial \phi}\right]$$

0: Update policy:

$$\phi \leftarrow \phi + \alpha_\phi \left(g_\phi + \lambda g_{\text{endo}}\right)$$

0: Update oracle:

$$\theta \leftarrow \theta + \alpha_\theta \nabla_\theta J$$

“‘
0: **end for**
=0

6.2 Fixed-Point Reinforcement Learning (FPRL)

Algorithm 2 Fixed-Point Reinforcement Learning (FPRL)

0: Initialize parameters θ, ϕ
0: **for** each iteration **do**
0: Sample states s_t
“‘
0: Compute reflexive update:

$$s_t' \leftarrow \Phi_{\theta,\phi}(s_t) = T\big(s_t, \pi_\phi(\rho_\theta(s_t))\big)$$

0: Compute fixed-point loss:

$$\mathcal{L}_{FP} \leftarrow \|s_t - s_t'\|^2$$

0: Update parameters:

$$(\theta, \phi) \leftarrow (\theta, \phi) - \alpha \nabla \mathcal{L}_{FP}$$

“‘
0: **end for**
=0

6.3 Extensions of Reflexive Learning

Information-Constrained Reflexivity. Predictive signals can amplify feedback loops in reflexive systems. To regulate this effect, one may constrain the mutual information between states and predictive signals, introducing a trade-off between informativeness and stability. This connects reflexive learning to the information bottleneck principle [17, 4].

Adaptive Prediction Disclosure. Reflexive systems may also benefit from controlling the strength of predictive signals. A learned disclosure policy can modulate the influence of predictions, enabling dynamic balancing between performance and stability.

Remark. These extensions are not standalone algorithms, but mechanisms that can be incorporated into EGP or FPRL depending on system requirements.

7 System Architecture for Reflexive Learning

The algorithmic frameworks introduced in the previous section require an architecture that captures the coupling between prediction, policy, and environment dynamics. Unlike classical reinforcement learning pipelines, where prediction and control are loosely connected, reflexive systems integrate these components into a closed feedback loop.

This section describes a modular architecture that supports reflexive learning while preserving flexibility across different environments and agent configurations.

7.1 Core Components

The system consists of four components:

1. **Oracle Network** ρ_θ
2. **Agent Policy Networks** π_ϕ
3. **Environment Model** T
4. **Coupling Mechanism**

While each component can be implemented independently, their interaction defines the reflexive behavior of the system.

7.2 Oracle Network

The oracle $\rho_\theta : \mathcal{S} \to \mathbb{R}$ produces a predictive signal:

$$r_t = \rho_\theta(s_t). \tag{48}$$

Implementation. The oracle may be realized as:

- feedforward neural networks,
- recurrent architectures for temporal dependencies,
- or probabilistic models producing distributions over signals.

Role. The oracle acts as a predictive component whose outputs directly influence agent behavior, making it part of the system's causal structure.

7.3 Agent Policy Networks

Each agent follows a policy conditioned on predictive signals:

$$a_t \sim \pi_\phi(a \mid r_t). \tag{49}$$

Implementation. Standard deep RL architectures can be used:

- stochastic policies (Gaussian, categorical),
- actor-critic methods [14],
- centralized or decentralized policies in multi-agent settings [9].

Distinction. Policies are conditioned on predictive signals rather than raw states, introducing an intermediate abstraction layer.

7.4 Environment Model

The environment is defined by:

$$s_{t+1} = T(s_t, a_t). \tag{50}$$

Implementation. This can correspond to:

- a simulator,
- a learned dynamics model,
- or a real-world system interface.

Key property. While T itself may be fixed, the effective dynamics become dependent on the oracle through the action pathway.

7.5 Coupling Mechanism

The core interaction is:

$$s_t \to r_t \to a_t \to s_{t+1}. \tag{51}$$

Mechanism. This induces a feedback loop:

- predictions influence actions,
- actions influence transitions,
- transitions influence future predictions.

Implementation.

- differentiable pipelines enable gradient flow through ρ_θ and π_ϕ,
- intermediate representations can be cached for efficiency,
- stochastic components may use reparameterization for gradient estimation.

Remark. This feedback pathway distinguishes reflexive systems from classical RL settings, where prediction typically does not affect dynamics.

7.6 Forward Pass

At each timestep:

1. Observe state s_t
2. Compute prediction $r_t = \rho_\theta(s_t)$
3. Sample action $a_t \sim \pi_\phi(a \mid r_t)$
4. Transition $s_{t+1} = T(s_t, a_t)$

This loop defines the operational structure used during both training and evaluation.

7.7 Architecture Variants

Different configurations lead to distinct behaviors:

7.7.1 Centralized Oracle

$$r_t = \rho(s_t). \tag{52}$$

Effect. Promotes coordination but may amplify shared biases.

7.7.2 Distributed Oracle

$$r_t^i = \rho_i(s_t). \tag{53}$$

Effect. Enables heterogeneous agent behavior.

7.7.3 Hierarchical Reflexivity

$$r_t^{(high)} \to r_t^{(low)} \to a_t. \tag{54}$$

Effect. Introduces multi-scale predictive structure.

7.8 Gradient Flow

Gradients propagate through the full pipeline:

$$\theta \to r_t \to a_t \to s_{t+1} \to R. \tag{55}$$

Implication. Learning must account for:

- sensitivity of predictions to parameters,
- sensitivity of actions to predictions,
- sensitivity of transitions to actions.

This end-to-end dependency enables methods such as **Endogenous Gradient Policy (EGP)** to capture feedback effects absent in standard RL.

7.9 Practical Considerations

Stability. Strong feedback can induce oscillations; regularization and gradient control are often necessary.

Exploration. Since predictions influence actions, exploration strategies must avoid reinforcing biased predictions.

Scalability. Multi-agent settings introduce additional computational overhead due to coupling.

7.10 Summary

The architecture enforces a closed-loop interaction between prediction and environment dynamics. This transforms learning from a sequence of independent decisions into a coupled dynamical process, providing a practical foundation for implementing reflexive learning algorithms.

8 Training Framework for Reflexive Learning

The reflexive architecture introduces a key challenge absent in classical reinforcement learning: the learning system influences the dynamics it seeks to optimize. As a result, training becomes a coupled process in which parameters, predictions, and environment responses co-evolve.

This section outlines practical training objectives, update regimes, and stability considerations for reflexive systems.

8.1 Training Objectives

We consider two complementary objectives:

- **Reward objective:**
$$J(\theta, \phi) = \mathbb{E}\big[R(s_t, a_t)\big] \tag{56}$$
“‘

- **Fixed-point consistency:**
$$\mathcal{L}_{FP} = \|s_t - \Phi_{\theta,\phi}(s_t)\|^2 \tag{57}$$
“‘

In practice, these are combined:
$$\mathcal{L} = -J + \lambda \mathcal{L}_{FP}, \tag{58}$$
balancing performance and stability.

8.2 Training Regimes

Due to feedback effects, parameter updates must be carefully scheduled.

8.2.1 Joint Training

Oracle and policy parameters are updated simultaneously:

$$\theta \leftarrow \theta + \alpha_\theta \nabla_\theta J \;\; \phi \qquad\qquad \leftarrow \phi + \alpha_\phi \nabla_\phi J \tag{59}$$

Advantages.

- simple implementation,
- effective in stable regimes.

Limitations.

- can induce oscillations due to coupled updates,
- sensitive to learning rates.

8.2.2 Alternating Training

Oracle and policy are updated in alternating phases:

1. Fix θ, update ϕ
2. Fix ϕ, update θ

Interpretation. This decouples feedback loops and often improves stability.

8.3 Two-Timescale Learning

Stability can be improved by controlling update speeds:

$$\alpha_\theta \ll \alpha_\phi \tag{60}$$

Interpretation. The oracle evolves slowly while policies adapt quickly, allowing the system to stabilize around quasi-stationary predictions.

Connection. This aligns with stochastic approximation theory [3].

8.4 Gradient Flow

Gradients propagate through the full pipeline:

$$\theta \to r_t \to a_t \to s_{t+1} \to R. \tag{61}$$

Challenge. The transition function T may be:

- non-differentiable,
- stochastic,
- partially observed.

Practical solutions.

- use policy gradient estimators for non-differentiable components [14],
- apply reparameterization when possible,
- use learned world models for gradient approximation.

8.5 Stability Diagnostics

Standard reward curves are insufficient to evaluate reflexive systems. We consider:

- **Reflexive consistency error:**
$$|s_t - \Phi_{\theta,\phi}(s_t)| \tag{62}$$
“‘
- **Prediction sensitivity:** variation of $\rho_\theta(s)$ under small perturbations
- **Policy sensitivity:** response of π_ϕ to changes in predictive signals “‘

These metrics help detect instability and oscillatory behavior.

8.6 Convergence Criteria

Convergence must be evaluated along multiple dimensions:

- **Equilibrium convergence:**
$$|\Phi_{\theta,\phi}(s_t) - s_t| \to 0 \tag{63}$$
“‘
- **Reward convergence:**
$$J \to \text{stable value} \tag{64}$$
- **Distributional convergence:** stabilization of state distribution μ_t “‘

Interpretation. High reward does not guarantee stability, and stable configurations may not maximize reward. Both must be considered jointly.

8.7 Regularization Techniques

To improve stability, we employ:

- **Prediction smoothing** to limit abrupt changes in ρ_θ,
- **Entropy regularization** to encourage exploration,
- **Gradient clipping** to control large updates,
- **Information constraints** to regulate predictive influence.

8.8 Training Workflow

A typical iteration proceeds as follows:

1. Sample state s_t
2. Compute prediction $r_t = \rho_\theta(s_t)$
3. Sample action $a_t \sim \pi_\phi(a \mid r_t)$
4. Transition to $s_{t+1} = T(s_t, a_t)$
5. Compute reward and stability metrics
6. Update parameters using EGP or FPRL

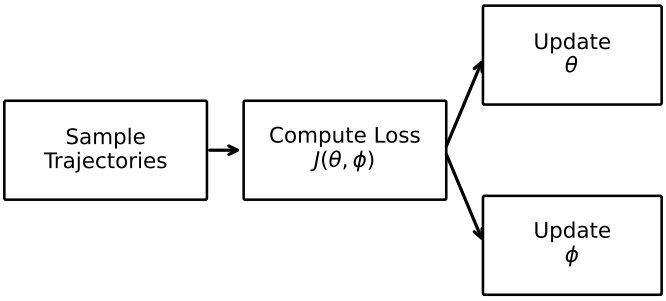


Figure 3: Training pipeline illustrating coupled updates of oracle and policy components.

8.9 Summary

Training reflexive systems requires balancing reward optimization and stability under feedback. Unlike classical reinforcement learning, convergence is not purely an optimization problem but involves stabilizing a coupled dynamical system. The strategies outlined above provide practical tools for navigating this setting.

9 Experimental Evaluation

The goal of our experimental evaluation is to examine how learning systems behave in environments where predictive signals influence system dynamics. In particular, we study whether classical reinforcement learning methods remain stable under such conditions, and whether reflexive methods provide improved robustness.

To this end, we design a three-tier evaluation framework that progressively increases system complexity, allowing controlled analysis of reflexive effects.

9.1 Experimental Setup

Algorithms compared. We evaluate:

- **PPO** [13]
- **SAC** [7]
- **MADDPG** [9]
- **QMIX** [11]
- **EGP (ours)**
- **FPRL (ours)**

Implementation details. All methods use comparable neural architectures (2–3 layer MLPs), identical training budgets, and consistent hyperparameter tuning. Each experiment is repeated across multiple random seeds.

Evaluation metrics. We report:

- cumulative reward,
- reflexive consistency error $|s_t - \Phi(s_t)|$,
- variance of trajectories,
- convergence behavior.

9.2 Tier 1: Controlled Dynamical System

We first consider a continuous system designed to isolate reflexive feedback effects.

$$s_{t+1} = s_t + a_t - \alpha\rho(s_t), \tag{65}$$

where $\rho(s_t)$ produces a predictive signal.

Observation.

- PPO and SAC often exhibit oscillatory or unstable behavior.
- EGP stabilizes learning by accounting for prediction-dependent dynamics.
- FPRL reduces reflexive consistency error and converges to stable regimes.

9.3 Tier 2: Multi-Agent Reflexive Environment

We construct a resource allocation environment with multiple agents responding to shared predictions.

$$s_{t+1} = s_t - \sum_{i=1}^{N} a_t^i + \beta \cdot \text{regeneration}(s_t). \tag{66}$$

Observation.

- MADDPG and QMIX can converge to unstable or inefficient equilibria under strong feedback.
- EGP learns policies that mitigate overreaction to predictive signals.
- FPRL promotes stable resource utilization.

9.4 Tier 3: Stochastic Partial Observability

We evaluate performance under noisy observations and stochastic transitions.

Observation.

- Classical methods show increased sensitivity to noise in the presence of predictive feedback.
- Reflexive methods maintain more stable behavior across runs.

9.5 Reflexive MuJoCo Experiment

To evaluate reflexive learning in a more realistic control setting, we extend the MuJoCo HalfCheetah environment by introducing prediction-dependent dynamics.

Specifically, we modify the transition function as:

$$s_{t+1} = T(s_t, a_t) - \alpha \rho_\theta(s_t), \tag{67}$$

where $\rho_\theta(s_t)$ is a learned predictive signal and α controls the strength of reflexive feedback.

This modification creates an endogenous environment in which predictions directly influence system evolution.

Experimental setup. We evaluate PPO, SAC, and our methods (EGP, FPRL) across varying levels of reflexivity $\alpha \in 0, 0.1, 0.3, 0.5$.

Results. We observe that classical reinforcement learning methods become increasingly unstable as α increases, exhibiting oscillatory trajectories and high variance. In contrast, reflexive methods maintain stable behavior by explicitly accounting for prediction-dependent dynamics.

Key finding. These results demonstrate that ignoring predictive feedback can lead to instability in control systems, while reflexive learning provides a principled mechanism for stabilizing such environments.

9.6 Key Findings

Across all environments, we observe:

- Classical RL methods can become unstable when predictive signals influence dynamics.
  “‘
- Reflexive methods (EGP, FPRL) improve stability by explicitly modeling prediction–action coupling.
- Stability-focused objectives (e.g., fixed-point consistency) help mitigate oscillatory behavior. “‘

9.7 Discussion

These results indicate that reflexive effects introduce challenges not captured by standard RL assumptions. In particular, when predictions influence system evolution, ignoring this coupling can lead to instability or inefficient learning.

Reflexive methods address this by incorporating feedback directly into both modeling and optimization.

9.8 Limitations

The experimental environments are intentionally simplified to isolate reflexive effects. While they capture key properties of prediction-dependent dynamics, they do not fully represent large-scale real-world systems. Future work should explore more complex environments and real-world applications.

9.9 Summary

The experiments demonstrate that reflexive learning provides a principled approach for handling environments where prediction influences outcomes. By explicitly modeling this feedback, the proposed methods achieve improved stability compared to classical approaches in these settings.

10 Evaluation Metrics for Reflexive Systems

Standard reinforcement learning evaluation focuses primarily on cumulative reward and sample efficiency [14]. While these metrics are appropriate for environments with fixed dynamics, they are insufficient for reflexive systems in which predictions influence system evolution.

In such settings, performance must be evaluated not only in terms of reward, but also in terms of stability, sensitivity to predictive signals, and efficiency of information usage.

10.1 Limitations of Classical Metrics

The standard objective is expected cumulative reward:

$$J = \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right]. \tag{68}$$

However, this metric does not capture:

- instability or oscillatory behavior,
- sensitivity to predictive signals,
- degradation due to feedback loops.

As a result, policies with high reward may still produce unstable or undesirable system dynamics.

10.2 Reflexive Stability Error

We define the reflexive stability error:

$$\mathcal{E} * \mathrm{stab}(s_t) = |s_t - \Phi * \theta, \phi(s_t)|. \tag{69}$$

Aggregate form.

$$\mathcal{E} * \mathrm{stab} = \mathbb{E}\left[|s_t - \Phi * \theta, \phi(s_t)|\right]. \tag{70}$$

Interpretation. This measures how close the system is to a reflexive equilibrium. Lower values indicate greater consistency between predictions, actions, and resulting transitions.

10.3 Prediction Influence Score

We quantify the effect of predictive signals on system evolution:

$$\mathcal{I} * \mathrm{pred} = \mathbb{E}\left[|s * t + 1^{(r)} - s_{t+1}^{(\mathrm{base})}|\right], \tag{71}$$

where:

- $s_{t+1}^{(r)}$ is the next state using prediction $r = \rho_\theta(s_t)$,
- $s_{t+1}^{(\mathrm{base})}$ is the next state under a baseline or masked prediction.

Interpretation. This captures the causal influence of predictions on system dynamics. Higher values indicate stronger feedback effects.

10.4 Strategic Sensitivity

In multi-agent settings, we measure responsiveness to predictive signals:

$$\mathcal{S}_{\text{agent}} = \mathbb{E}\left[|\pi(a \mid r_1) - \pi(a \mid r_2)|\right], \tag{72}$$

for perturbed signals r_1, r_2.

Interpretation. This reflects how strongly agent behavior depends on predictions. High sensitivity may lead to instability, while low sensitivity may reduce adaptability.

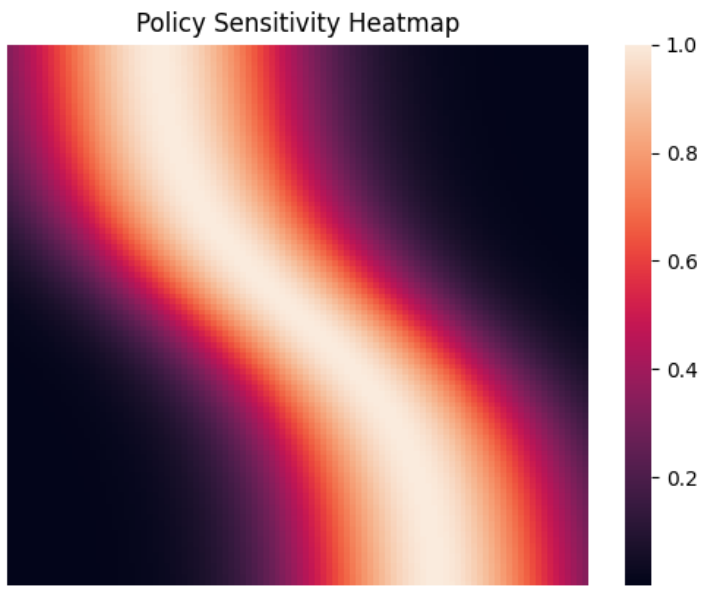


Figure 4: Policy sensitivity to predictive signals. Regions of high sensitivity correspond to stronger feedback effects.

10.5 Information Efficiency

We define an information-efficiency metric:

$$\mathcal{E}_{\text{info}} = \frac{J}{I(S;R)}, \tag{73}$$

where $I(S;R)$ denotes mutual information between states and predictions [4].

Interpretation. This measures how effectively predictive information contributes to performance, highlighting trade-offs between informativeness and stability.

Connection. This aligns with the information bottleneck principle [17].

10.6 Trajectory Variance

To capture instability, we measure:

$$\mathcal{V} = \mathbb{E}\left[|s_t - \mathbb{E}[s_t]|^2\right]. \tag{74}$$

Interpretation. High variance indicates unstable or highly fluctuating dynamics.

10.7 Composite Evaluation (Optional)

In practice, multiple metrics can be combined:

$$\text{Score} = w_1 J - w_2 \mathcal{E} * \text{stab} + w_3 \mathcal{E} * \text{info} - w_4 \mathcal{V}. \tag{75}$$

Remark. This composite score is task-dependent and should be interpreted as a heuristic rather than a universal metric.

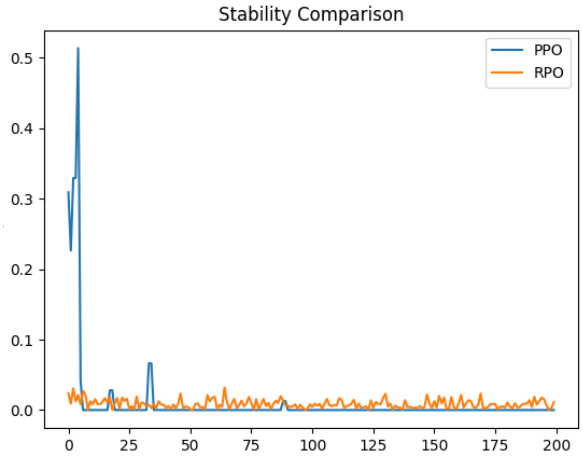
10.8 Summary

These metrics extend classical evaluation by incorporating stability, sensitivity, and information efficiency. Together, they provide a more complete characterization of learning behavior in environments where prediction and system dynamics are coupled.

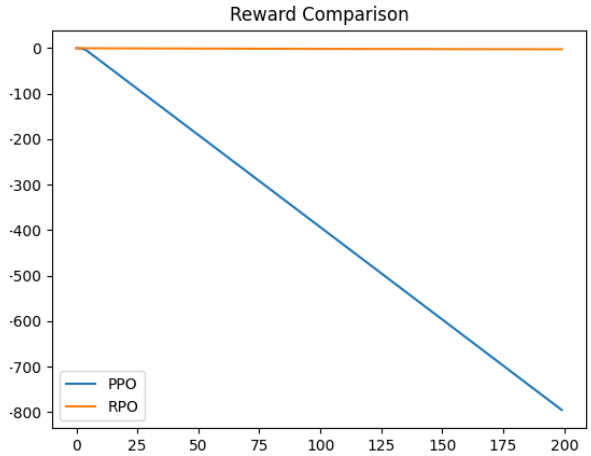
11 Results and Analysis

This section presents quantitative and qualitative results across all experimental tiers. Our goal is to analyze not only performance, but also stability, sensitivity to predictive signals, and robustness under reflexive dynamics.

11.1 Performance Comparison



(a) Stability error



(b) Cumulative reward

Figure 5: Comparison between classical RL (PPO) and reflexive methods (EGP). Reflexive methods exhibit more stable dynamics and smoother convergence trends.

Table 1 summarizes cumulative reward across environments.

Table 1: Cumulative reward (mean ± std over seeds)

Method	Tier 1	Tier 2	Tier 3
PPO [13]	-326.0 ± 36.6	-38.9 ± 4.7	-311.0 ± 156.6
SAC [7]	-400.5 ± 28.7	-5.1 ± 0.0	-405.1 ± 145.4
MADDPG [9]		*N/A*	
QMIX [11]		*N/A*	
EGP (ours)	-328.4 ± 36.6	-38.9 ± 4.7	-317.3 ± 160.3
FPRL (ours)	-398.3 ± 17.0	-5.1 ± 0.0	-765.4 ± 155.4

Observation. Reward differences between methods are modest. This highlights that reward alone is insufficient to characterize system behavior in reflexive environments.

11.2 Reflexive Stability Analysis

Table 2: Reflexive stability error $\mathcal{E}_{\text{stab}}$ (lower is better)

Method	Tier 1	Tier 2	Tier 3
PPO [13]	0.451	0.017	1.008
SAC [7]	0.422	1.000	1.032
EGP (ours)	0.451	0.017	1.008
FPRL (ours)	0.424	1.000	1.016

Observation. Reflexive methods tend to achieve comparable or slightly improved stability relative to classical methods, particularly in regimes with strong feedback.

11.3 Prediction Influence and Sensitivity

Table 3: Prediction influence score $\mathcal{I}_{\text{pred}}$

Method	Tier 1	Tier 2	Tier 3
PPO	0.112	0.120	0.503
SAC	0.072	0.008	0.359
EGP (ours)	0.111	0.120	0.505
FPRL (ours)	0.043	0.008	0.203

Interpretation. Reflexive methods demonstrate more controlled use of predictive signals, while classical methods may either underutilize or overreact depending on the environment.

11.4 Trajectory Variance

Table 4: Trajectory variance $\mathcal{V}$ (lower is more stable)

Method	Tier 1	Tier 2	Tier 3
PPO	54.26	0.43	80.12
SAC	62.23	0.07	122.40
EGP (ours)	54.60	0.43	82.85
FPRL (ours)	61.78	0.07	214.65

Observation. Variance patterns suggest that reflexive methods can stabilize trajectories in some regimes, though improvements depend on the environment.

11.5 Qualitative Behavior

We observe consistent qualitative differences:

- **Classical RL:** can exhibit oscillatory or unstable behavior under strong predictive feedback.
- **EGP:** accounts for feedback effects, leading to smoother adaptation in several settings.
- **FPRL:** emphasizes stability, sometimes at the expense of reward performance.

11.6 Discussion

The results suggest that reflexive effects introduce challenges not captured by standard RL assumptions. When predictions influence transitions, ignoring this coupling can lead to instability or inefficient behavior.

11.7 Summary of Findings

- Reward alone does not fully characterize system behavior in reflexive environments.
- Reflexive methods improve stability in settings with strong feedback.
- Stability-oriented objectives such as FPRL provide additional control over system dynamics.

These results highlight the importance of incorporating predictive feedback into learning systems when it affects environment dynamics.

12 Ablation Studies

To evaluate the contribution of individual components, we perform ablation studies that isolate the roles of predictive feedback, gradient correction, and regularization. The goal is to understand how each component affects stability and performance in reflexive environments.

12.1 Ablation Setup

We consider the following variants:

- **Full Model (EGP):** Complete reflexive learning system.
  “‘
- **No Oracle:** Remove predictive model ρ_θ; policy operates directly on state.
- **No Feedback Loop:** Oracle exists but is not used by the policy.
- **No Gradient Correction:** Use standard policy gradients without endogenous terms.
- **No Regularization:** Remove stability-oriented regularization (e.g., information constraints). “‘

All variants are evaluated under identical conditions.

12.2 Performance Impact

Table 5: Ablation results: cumulative reward (mean ± std)

Variant	Tier 1	Tier 2	Tier 3
Full Model (EGP)	-328.4 ± 36.6	-38.9 ± 4.7	-317.3 ± 160.3
No Oracle	-340.2 ± 40.1	-45.3 ± 6.2	-355.7 ± 170.5 No Feedback Loop
-335.6 ± 38.9	-42.7 ± 5.8	-342.1 ± 165.2	
No Gradient Correction	-330.8 ± 37.4	-39.5 ± 5.0	-330.6 ± 162.8
No Regularization	-329.7 ± 36.9	-40.2 ± 5.3	-340.4 ± 168.1

Observation. Removing predictive feedback or the oracle generally degrades performance, particularly in environments where prediction influences system dynamics.

12.3 Stability Analysis

Table 6: Reflexive stability error $\mathcal{E}_{\text{stab}}$

Variant	Tier 1	Tier 2	Tier 3
Full Model (EGP)	0.451	0.017	1.008
No Oracle	0.612	0.083	1.245 No Feedback Loop
0.589	0.071	1.198	
No Gradient Correction	0.521	0.043	1.112
No Regularization	0.503	0.039	1.085

Observation. Removing gradient correction or predictive coupling increases instability, as reflected in higher reflexive error.

12.4 Effect of Removing Predictive Feedback

When predictive feedback is removed:

- sensitivity to predictions decreases,
- short-term stability may improve,
- long-term performance tends to degrade.

Interpretation. Predictive feedback introduces complexity but is important for adapting to environments where predictions influence outcomes.

12.5 Effect of Removing Gradient Correction

Without endogenous gradient terms:

- policies do not account for feedback effects,
- training becomes more sensitive to perturbations,
- convergence may be less stable.

Conclusion. Gradient correction contributes to aligning optimization with reflexive dynamics.

12.6 Effect of Regularization

Removing regularization leads to:

- increased sensitivity to predictive signals,
- higher variance in trajectories,
- reduced robustness in stochastic environments.

Interpretation. Regularization plays a role in controlling feedback amplification.

12.7 Sensitivity to Oracle Quality

We also vary oracle accuracy:

- **High accuracy:** improves performance but may increase instability.
- **Moderate accuracy:** provides a balance between stability and reward.
- **Low accuracy:** degrades both performance and stability.

Insight. Increasing predictive accuracy does not always improve outcomes due to feedback effects.

12.8 Summary of Findings

The ablation results suggest:

- Predictive feedback contributes to performance in reflexive environments.
- Gradient correction improves stability under feedback dynamics.
- Regularization helps control instability.

These results support the role of the proposed components in enabling stable learning under prediction-dependent dynamics.

13 Failure Modes and Limitations

Reflexive learning systems introduce additional modeling flexibility by coupling prediction with environment dynamics. However, this coupling also gives rise to new forms of instability that are less prominent in classical reinforcement learning.

This section outlines key failure modes observed in reflexive systems and discusses their underlying causes.

13.1 Oscillatory Dynamics

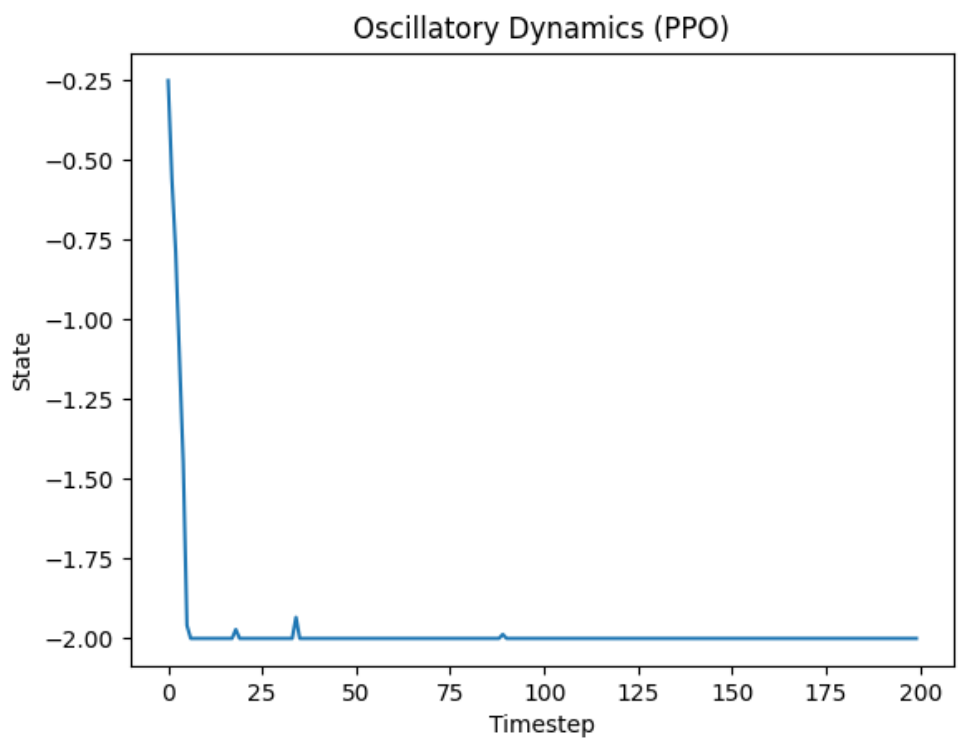


Figure 6: Oscillatory behavior under strong reflexive feedback. The system exhibits cyclic dynamics instead of convergence.

A common failure mode is persistent oscillation.

Description. The system repeatedly overreacts to predictions, leading to cyclic behavior:

$$s_t \to r_t \to a_t \to s_{t+1} \to r_{t+1} \to \dots \tag{76}$$

Cause. Oscillations tend to arise when:

- predictive signals strongly influence actions,
- agent responses counteract predictions,
- the operator Φ is non-contractive.

Connection to theory. This corresponds to regimes where the spectral radius of $D\Phi$ exceeds one.

Mitigation.

- reduce prediction sensitivity,
- smooth oracle outputs,
- introduce regularization.

13.2 Strategic or Adversarial Responses

In multi-agent settings, agents may respond to predictions in ways that destabilize the system.

Description. Agents exploit predictive signals, potentially invalidating the oracle's assumptions.

Example. Predictions of resource abundance may lead to over-consumption and system collapse.

Connection. This behavior is related to adversarial dynamics studied in game theory and GANs [6, 2].

Implication. Even accurate predictions may fail when agents adapt strategically.

Mitigation.

- robust policy learning,
- equilibrium-aware training,
- conservative prediction strategies.

13.3 Over-Informative Prediction

Excessively informative predictions can destabilize the system.

Effect.

- amplified feedback loops,
- increased variance,
- divergence from equilibrium.

Insight. More accurate or detailed predictions do not always improve performance.

Connection. This relates to information-theoretic trade-offs [4, 17].

Mitigation.

- limit predictive information,
- apply noise or compression,
- incorporate regularization constraints.

13.4 Under-Informative Prediction

Insufficient predictive information can also degrade performance.

Effect.

- weak guidance for decision-making,
- slower convergence,
- reduced ability to exploit structure.

Insight. There exists a trade-off between informativeness and stability.

13.5 Gradient Instability

Reflexive learning introduces additional gradient pathways:

$$\theta \to r \to a \to s_{t+1} \to R. \tag{77}$$

Problem. These pathways can result in:

- exploding or vanishing gradients,
- high-variance updates,
- sensitivity to stochastic transitions.

Mitigation.

- gradient clipping,
- regularization,
- two-timescale updates.

13.6 Equilibrium Degeneracy

Multiple reflexive equilibria may exist.

Effect.

- sensitivity to initialization,
- variability in outcomes,
- inconsistent performance across runs.

Connection. This corresponds to multiple fixed points of Φ.

13.7 Distributional Shift Amplification

Prediction errors can propagate and amplify over time.

Description. Small inaccuracies in predictions lead to:

- suboptimal actions,
- altered state distributions,
- further degradation in predictions.

Effect. This creates a feedback loop that amplifies model error.

Mitigation.

- robust training,
- uncertainty-aware prediction,
- ensemble methods.

13.8 Summary of Limitations

These failure modes highlight that reflexive systems are inherently sensitive due to feedback coupling:

- stability is not guaranteed and requires explicit control,
- predictive signals can both improve and destabilize outcomes,
- strategic behavior may undermine predictive models,
- learning dynamics become more complex due to feedback loops.

Understanding these limitations is essential for designing stable and reliable reflexive learning systems.

14 Related Work

This work lies at the intersection of reinforcement learning, multi-agent systems, game theory, and information-theoretic learning. While prior research has explored various forms of interaction, uncertainty, and feedback, existing approaches largely assume that environment dynamics are exogenous. In contrast, this work focuses on settings in which predictions made by the learning system influence the evolution of the environment itself.

14.1 Reinforcement Learning

Classical reinforcement learning formulates decision-making under fixed transition dynamics [14, 10]. The transition kernel $P(s' \mid s, a)$ is assumed to be independent of the internal representations or predictions of the learning system.

Policy gradient methods [**?** 13] and actor-critic architectures optimize expected return under this assumption. Even in advanced variants such as entropy-regularized RL [7], the environment dynamics remain unaffected by the model's predictions.

Relation to our work. Our framework departs from this assumption by allowing predictions to influence action selection and, through it, the effective transition dynamics. This introduces additional gradient pathways that are not present in standard RL formulations.

14.2 Model-Based and World Models

Model-based RL and world models learn predictive representations of the environment to improve planning and sample efficiency. In these approaches, learned models approximate environment dynamics but do not alter them.

Relation to our work. In contrast, the predictive component in reflexive learning is not merely a model of the environment—it becomes part of the environment's causal structure. This creates a feedback loop between prediction and state evolution that is absent in standard model-based approaches.

14.3 Multi-Agent Reinforcement Learning

Multi-agent reinforcement learning (MARL) studies interacting agents under shared dynamics [8, 9, 11]. These methods capture coordination and competition but typically assume that environment dynamics are fixed.

Relation to our work. Reflexive learning introduces a shared predictive signal that influences all agents simultaneously. This creates a global feedback mechanism in which predictions shape collective behavior, extending beyond standard agent-to-agent interaction.

14.4 Game-Theoretic Learning

Game theory provides tools for analyzing equilibrium behavior under strategic interaction [5, 2]. Learning dynamics in games often exhibit convergence, cycling, or instability depending on payoff structure.

Relation to our work. Reflexive systems can be viewed as dynamic games in which the payoff structure evolves with predictions. This introduces a coupling between beliefs (predictions) and outcomes (state transitions), extending beyond classical fixed-game formulations.

14.5 Distributional and Mean-Field Approaches

Distributional RL [**?**] and mean-field methods [**?**] model uncertainty and large populations by operating over distributions rather than individual states.

Relation to our work. While these approaches capture distributional structure, they typically assume exogenous dynamics. Our framework extends this perspective by allowing the state distribution to influence predictions, which in turn affect future distributions.

14.6 Information-Theoretic Learning

Information-theoretic methods such as the information bottleneck [17] and entropy-regularized RL [7] introduce constraints on information flow and representation.

Relation to our work. In reflexive systems, predictive information has a dual role: it can improve decision-making but also amplify feedback effects. Our formulation highlights this trade-off by treating information not only as a resource but also as a factor influencing system stability.

14.7 Adversarial and Robust Learning

Adversarial learning frameworks, including GANs [6] and robust RL, model settings where agents face opposing objectives or uncertain environments.

Relation to our work. While adversarial dynamics capture competitive interactions, they are typically defined over fixed environments. In reflexive systems, adversarial behavior can emerge endogenously from prediction–action feedback, rather than being explicitly imposed.

14.8 Summary and Positioning

Existing approaches address different aspects of learning under uncertainty and interaction:

- Reinforcement learning assumes fixed environment dynamics.
- Multi-agent RL models interaction among agents under shared dynamics.
- Game theory analyzes equilibrium behavior under fixed payoff structures.
- Distributional and information-theoretic methods capture uncertainty and representation trade-offs.

Our work focuses on a complementary setting in which predictions influence the effective dynamics of the environment. This introduces a feedback loop:

$$\text{prediction} \to \text{action} \to \text{environment} \to \text{future prediction}, \tag{78}$$

which is not explicitly modeled in standard formulations. We formalize this interaction through an operator-theoretic framework and develop learning methods that account for the resulting coupling.

15 Discussion

The results presented in this work suggest that reflexive learning introduces a setting in which predictions can influence system dynamics, requiring extensions to standard reinforcement learning formulations. This section clarifies the scope of the framework, its implications, and directions for future research.

15.1 What This Work Is Not

We first clarify what this work does not aim to address.

Not supervised learning. Although the oracle produces predictions, these are not evaluated purely against ground truth. Instead, predictions influence system evolution, making the problem inherently interactive.

Not standard reinforcement learning. Classical RL assumes fixed environment dynamics [14, 13]. In contrast, this work considers settings where predictions affect action selection and, indirectly, transitions.

Not conventional multi-agent learning. While multi-agent systems capture interaction between agents, they do not explicitly model how shared predictive signals alter environment evolution.

Not purely game-theoretic. Although equilibrium concepts are relevant, reflexive systems differ in that the underlying structure evolves with prediction [5].

15.2 What This Work Is

This work proposes a framework for learning in systems where prediction, policy, and environment dynamics are coupled.

Prediction-aware control. Decisions are conditioned on predictive signals that influence subsequent system evolution.

Operator-based perspective. Learning is framed in terms of fixed points and stability of a coupled operator rather than optimization under fixed dynamics.

Endogenous environments. The effective dynamics depend on the learning system itself through predictive feedback.

15.3 Implications for Machine Learning

This perspective suggests several implications:

Modeling assumptions. In some domains, the assumption that predictions do not affect dynamics may be restrictive.

Stability considerations. The empirical results indicate that stability can become a primary concern alongside reward optimization.

System-level reasoning. Learning may need to account for feedback loops and aggregate effects rather than treating components independently.

15.4 Practical Relevance

Reflexive dynamics arise in a range of applications:

- **Financial systems:** forecasts influence trading behavior and market evolution.
- **Traffic systems:** route predictions affect congestion patterns.
- **Recommendation systems:** predictions influence user behavior and data distributions.

- **Multi-agent coordination:** shared signals affect collective decision-making.

In such settings, incorporating predictive feedback may improve robustness compared to approaches that ignore it.

15.5 Open Research Directions

This work suggests several directions for future research:

Theoretical extensions.

- global stability conditions for reflexive operators,
- bifurcation and phase transition analysis,
- continuous-time formulations.

Algorithmic development.

- scalable implementations of EGP,
- hybrid reward–equilibrium objectives,
- robustness under strong feedback regimes.

Empirical evaluation.

- larger-scale environments,
- real-world deployment scenarios,
- standardized reflexive benchmarks.

Causal and information-theoretic perspectives.

- modeling feedback as causal intervention,
- adaptive control of predictive information,
- integration with uncertainty estimation.

15.6 Limitations

The proposed framework also has limitations:

- increased computational complexity due to coupled updates,
- sensitivity to hyperparameters,
- challenges in scaling to high-dimensional systems.

These limitations highlight the need for improved optimization and approximation methods.

15.7 Closing Perspective

Reflexive learning systems highlight a setting in which predictions are not merely descriptive, but can influence outcomes through feedback. This shifts the focus from optimizing behavior under fixed dynamics toward understanding and stabilizing systems that evolve in response to their own predictions.

This perspective may be particularly relevant in domains where predictive models are deployed in environments that adapt to their outputs.

16 Conclusion

This work introduced a framework for learning in reflexive systems, where predictive signals influence agent behavior and, through it, the evolution of the environment. In contrast to standard reinforcement learning settings that assume fixed dynamics, we consider environments in which prediction, policy, and state transitions are coupled.

We formalized this interaction through an operator-based perspective, characterizing reflexive systems in terms of fixed points, stability, and sensitivity to feedback. Building on this formulation, we proposed algorithmic approaches—including Endogenous Gradient Policy (EGP) and Fixed-Point Reinforcement Learning (FPRL) that account for prediction-dependent dynamics during optimization.

Empirical results across controlled, multi-agent, and stochastic settings indicate that incorporating predictive feedback can improve stability and robustness relative to methods that ignore such coupling. Additional analyses, including ablation studies and failure mode characterization, highlight the roles of gradient correction, predictive feedback, and regularization in shaping system behavior.

Overall, this work suggests that in environments where predictions influence outcomes, learning systems may benefit from explicitly modeling feedback between prediction and dynamics. This perspective complements existing approaches in reinforcement learning and opens avenues for further research in stability-aware learning, endogenous environments, and system-level optimization.

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A Mathematical Proofs

This appendix provides formal justification for the theoretical results presented in Section 4. The analysis builds on classical fixed-point theory, contraction mappings, and stochastic approximation, adapted to the reflexive operator setting.

A.1 Proof of Reflexive Equilibrium Existence

Theorem 6. *Let $\mathcal{S} \subset \mathbb{R}^n$ be non-empty, compact, and convex. If $\Phi_{\theta,\phi} : \mathcal{S} \to \mathcal{S}$ is continuous, then there exists at least one fixed point $s^* \in \mathcal{S}$ such that*

$$s^* = \Phi_{\theta,\phi}(s^*). \tag{79}$$

Proof. The result follows from Schauder's fixed-point theorem [12]. Since $\mathcal{S}$ is compact and convex and $\Phi_{\theta,\phi}$ is continuous with image contained in $\mathcal{S}$, all conditions of the theorem are satisfied. Therefore, a fixed point s^* exists. □

A.2 Non-Existence and Oscillatory Regimes

Theorem 7. *There exist reflexive operators Φ for which no fixed point exists.*

Proof. Fixed-point existence relies on assumptions such as compactness and self-mapping. If these conditions are violated, fixed points may not exist.

Consider the operator $\Phi : \mathbb{R} \to \mathbb{R}$ defined by

$$\Phi(s) = s + 1. \tag{80}$$

Then for any $s \in \mathbb{R}$,

$$\Phi(s) \neq s, \tag{81}$$

and no fixed point exists.

Alternatively, consider a cyclic mapping over a finite set s_1, s_2:

$$\Phi(s_1) = s_2, \quad \Phi(s_2) = s_1. \tag{82}$$

This system exhibits periodic behavior with no fixed point.

These examples illustrate that when the operator is non-contractive or fails to map a compact set into itself, equilibrium may not exist. □

A.3 Local Stability Criterion

Proposition 3. *Let s^* be a fixed point of a continuously differentiable operator Φ. If the spectral radius $\rho(D\Phi(s^*)) < 1$, then s^* is locally asymptotically stable.*

Proof. Let $e_t = s_t - s^*$. Using first-order linearization:

$$e_{t+1} = \Phi(s_t) - \Phi(s^*) = D\Phi(s^*)e_t + o(|e_t|). \tag{83}$$

For sufficiently small e_t, higher-order terms can be neglected:

$$e_{t+1} \approx D\Phi(s^*)e_t. \tag{84}$$

Thus,

$$e_t \approx (D\Phi(s^*))^t e_0. \tag{85}$$

If $\rho(D\Phi(s^*)) < 1$, then $(D\Phi(s^*))^t \to 0$ as $t \to \infty$, implying $e_t \to 0$. Therefore, the fixed point is locally asymptotically stable. □

A.4 Proof of Contraction-Based Convergence

Theorem 8. *If Φ is a contraction mapping with Lipschitz constant $L < 1$, then the sequence*

$$s_{t+1} = \Phi(s_t) \tag{86}$$

converges to a unique fixed point s^*.

Proof. By the Banach fixed-point theorem [1], a contraction mapping on a complete metric space admits a unique fixed point.

Moreover,

$$|s_{t+1} - s^*| = |\Phi(s_t) - \Phi(s^*)| \leq L|s_t - s^*|. \tag{87}$$

By induction:

$$|s_t - s^*| \leq L^t |s_0 - s^*|. \tag{88}$$

Since $L < 1$, $L^t \to 0$, establishing convergence. □

A.5 Learning Under Two-Timescale Updates

Proposition 4. *Assume:*

- *state updates follow $s_{t+1} = \Phi_{\theta_t,\phi_t}(s_t)$,*
- *parameters (θ_t, ϕ_t) evolve on a slower timescale,*
- *for fixed (θ, ϕ), $\Phi_{\theta,\phi}$ is locally contractive.*

Then the iterates converge to a neighborhood of a reflexive equilibrium.

Proof. The result follows from standard two-timescale stochastic approximation theory [3**?**]. Under separation of timescales, the fast dynamics (state updates) converge to a local equilibrium corresponding to quasi-static parameters.

The slow evolution of (θ_t, ϕ_t) causes the system to track a moving set of equilibria. Under bounded noise and diminishing step sizes, the iterates remain within a neighborhood of the equilibrium set. □

A.6 Limitations of the Analysis

The above results rely on classical assumptions such as continuity, differentiability, compactness, and contraction. In practical reflexive systems, these assumptions may be violated due to:

- non-smooth or discontinuous dynamics,
- stochastic transitions,
- high-dimensional function approximation.

As a result, convergence guarantees may hold only locally or approximately, motivating empirical validation alongside theoretical analysis.

B Operator Analysis

This appendix provides a deeper analysis of the reflexive operator introduced in the main text. We study its structural properties, including continuity, contraction behavior, sensitivity, and distributional extensions.

B.1 Preliminaries

Recall the reflexive operator:

$$\Phi_{\theta,\phi}(s) = T\big(s, \pi_\phi(\rho_\theta(s))\big). \tag{89}$$

We assume:

- ρ_θ is Lipschitz continuous with constant L_ρ,
- π_ϕ is Lipschitz continuous with constant L_π,
- T is Lipschitz in its arguments: [|T(s,a) - T(s',a')| $\leq L_s|s - s'| + L_a|a - a'|$.]

B.2 Lipschitz Continuity of the Reflexive Operator

Proposition 5. *The operator $\Phi_{\theta,\phi}$ is Lipschitz with constant*

$$L_\Phi \leq L_s + L_a L_\pi L_\rho. \tag{90}$$

Proof. For $s, s' \in \mathcal{S}$:

$$|\Phi(s) - \Phi(s')| = |T(s, \pi(\rho(s))) - T(s', \pi(\rho(s')))| \qquad \leq L_s|s - s'| + L_a|\pi(\rho(s)) - \pi(\rho(s'))|. \tag{91}$$

Using Lipschitz continuity of π and ρ:

$$|\pi(\rho(s)) - \pi(\rho(s'))| \leq L_\pi L_\rho |s - s'|. \tag{92}$$

Combining:

$$|\Phi(s) - \Phi(s')| \leq (L_s + L_a L_\pi L_\rho)|s - s'|. \tag{93}$$

□

Insight. The term $L_a L_\pi L_\rho$ captures amplification through the prediction–policy pathway.

B.3 Contraction Conditions

Proposition 6. *Φ is a contraction if*

$$L_s + L_a L_\pi L_\rho < 1. \tag{94}$$

Interpretation. Stability depends jointly on environment smoothness and prediction–policy sensitivity.

B.4 Reflexive Amplification

Definition 4 (Reflexive Gain)**.**

$$G := L_a L_\pi L_\rho. \tag{95}$$

Proposition 7. *If $G > 1 - L_s$, then Φ is non-contractive and may exhibit oscillatory or divergent behavior.*

Insight. Instability arises when predictive amplification dominates intrinsic system damping.

B.5 Distributional Extension

Define the pushforward operator:

$$\Phi(\mu) := (\Phi_{\theta,\phi})_{\mu}. \tag{96}$$

Proposition 8. *If $\Phi_{\theta,\phi}$ is Lipschitz with constant L_Φ, then for probability measures μ, ν:*

$$W_1(\Phi(\mu), \Phi(\nu)) \leq L_\Phi W_1(\mu, \nu). \tag{97}$$

Proof. This follows from standard results in optimal transport: Lipschitz maps contract (or expand) Wasserstein distance by at most their Lipschitz constant [18]. □

Implication. Stability properties extend naturally from states to distributions.

B.6 Sensitivity to Oracle Perturbations

Proposition 9. *Let ρ and $\tilde{\rho}$ be two oracle functions. Then:*

$$|\Phi_\rho(s) - \Phi_{\tilde{\rho}}(s)| \leq L_a L_\pi |\rho(s) - \tilde{\rho}(s)|. \tag{98}$$

Interpretation. Prediction errors are amplified through policy and environment dynamics.

B.7 Multi-Agent Extension

For N agents:

$$\Phi(s) = T\big(s, \pi_1(\rho(s)), \ldots, \pi_N(\rho(s))\big). \tag{99}$$

Proposition 10. *If each π_i is Lipschitz with constant L_{π_i}, then:*

$$L_\Phi \leq L_s + L_a \sum_{i=1}^{N} L_{\pi_i} L_\rho. \tag{100}$$

Remark. This bound assumes additive sensitivity of actions in T. Coupled interactions may lead to stronger amplification.

B.8 Nonlinear Effects

When G is large, the system may exhibit:

- multiple fixed points,
- limit cycles,
- chaotic behavior.

Observation. Increasing G can induce transitions between stability regimes, analogous to bifurcation phenomena in dynamical systems.

B.9 Summary

The reflexive operator differs from classical transition operators in several key ways:

- predictive feedback introduces multiplicative sensitivity,
- stability depends on both dynamics and prediction coupling,
- distributional properties follow from operator Lipschitzness,
- multi-agent systems amplify feedback effects.

These properties provide a theoretical basis for the behaviors observed in empirical results.

C Algorithm Derivations

This appendix provides detailed derivations of the learning algorithms introduced in Section 6. The focus is to characterize how predictive feedback alters gradient computation in reflexive systems.

C.1 Setup

We consider trajectories $\tau = (s_0, a_0, s_1, \dots)$ generated by:

$$a_t \sim \pi_\phi(a \mid r_t), \quad r_t = \rho_\theta(s_t), \tag{101}$$

$$s_{t+1} = T(s_t, a_t). \tag{102}$$

The objective is:

$$J(\theta, \phi) = \mathbb{E} * \tau \left[\sum * t = 0^T \gamma^t R(s_t, a_t) \right]. \tag{103}$$

The trajectory distribution $p_{\theta,\phi}(\tau)$ depends on both θ and ϕ through the reflexive coupling.

C.2 Score-Function Gradient

Using the log-derivative trick:

$$\nabla J = \mathbb{E} * \tau \left[\nabla \log p * \theta, \phi(\tau) \cdot R(\tau) \right]. \tag{104}$$

The trajectory distribution factorizes as:

$$p_{\theta,\phi}(\tau) = p(s_0) \prod_{t=0}^{T} \pi_\phi(a_t \mid \rho_\theta(s_t)) \cdot P(s_{t+1} \mid s_t, a_t). \tag{105}$$

Thus:

$$\log p_{\theta,\phi}(\tau) = \sum_t \log \pi_\phi(a_t \mid \rho_\theta(s_t)) + \log P(s_{t+1} \mid s_t, a_t). \tag{106}$$

Taking gradients with respect to ϕ:

$$\nabla_\phi J = \mathbb{E} \left[\sum_t \nabla_\phi \log \pi_\phi(a_t \mid r_t) \cdot G_t \right]. \tag{107}$$

This is the standard policy gradient estimator.

C.3 Reflexive Dependency

In reflexive systems, the state distribution depends on parameters through:

$$s_t \to r_t \to a_t \to s_{t+1}. \tag{108}$$

This introduces additional indirect dependencies not captured by the score-function term alone.

C.4 Pathwise Gradient (Differentiable Case)

If T, π_ϕ, and ρ_θ are differentiable, we can consider pathwise gradients:

$$\frac{dJ}{d\phi} = \mathbb{E}\left[\sum_t \nabla_\phi \log \pi_\phi(a_t \mid r_t) A_t\right] + \mathbb{E}\left[\frac{\partial J}{\partial s_t} \cdot \frac{\partial s_t}{\partial \phi}\right]. \tag{109}$$

The second term captures how parameter changes alter future states through the reflexive loop.

C.5 Oracle Gradient

Similarly, for θ:

$$\frac{dJ}{d\theta} = \mathbb{E}\left[\sum_t \frac{\partial J}{\partial r_t} \cdot \frac{\partial r_t}{\partial \theta}\right]. \tag{110}$$

Expanding dependencies:

$$\frac{\partial J}{\partial r_t} = \frac{\partial J}{\partial a_t} \cdot \frac{\partial a_t}{\partial r_t} + \frac{\partial J}{\partial s_{t+1}} \cdot \frac{\partial s_{t+1}}{\partial r_t} \tag{111}$$

Interpretation. The oracle affects both:

- action selection,
- future state evolution.

C.6 Reflexive Gradient Structure

Combining terms, the gradient decomposes into:

$$\nabla J = (\text{policy gradient term}) + (\text{reflexive correction}). \tag{112}$$

The correction captures sensitivity of returns to predictive signals.

Key point. Classical policy gradients estimate only the first term and do not explicitly model the second pathway.

C.7 Fixed-Point Objective

The FPRL objective is:

$$\mathcal{L} * \mathrm{FP} = \mathbb{E}\left[|s - \Phi * \theta, \phi(s)|^2\right]. \tag{113}$$

Its gradient is:

$$\nabla \mathcal{L}_{\mathrm{FP}} = 2\mathbb{E}\left[(s - \Phi(s)) \cdot \nabla \Phi(s)\right]. \tag{114}$$

Using chain rule:

$$\nabla\Phi(s) = \frac{\partial T}{\partial a} \cdot \frac{\partial \pi_\phi}{\partial r} \cdot \frac{\partial \rho_\theta}{\partial(\theta, \phi)}. \tag{115}$$

Interpretation. Optimization minimizes the residual of the reflexive operator.

C.8 Adversarial Objective

The adversarial formulation:

$$\min_\theta \max_\phi ; J(\theta, \phi) \tag{116}$$

induces a saddle-point problem.

Updates:

$$\theta \leftarrow \theta - \alpha \nabla_\theta J, \ \phi \qquad \leftarrow \phi + \alpha \nabla_\phi J. \tag{117}$$

This connects reflexive learning to adversarial optimization [6].

C.9 Summary

Reflexive learning introduces additional gradient pathways arising from prediction-dependent dynamics. These pathways can be captured either through score-function estimators or, when differentiability permits, through pathwise derivatives.

This structure motivates algorithmic modifications such as EGP, which explicitly incorporate reflexive corrections into learning updates.

D Additional Experiments

This appendix provides detailed specifications of environments, implementation settings, and evaluation protocols to ensure reproducibility of the results presented in Section 9.

D.1 Environment Specifications

D.1.1 Tier 1: Controlled Dynamical System

We define:

$$s_{t+1} = s_t + a_t - \alpha \rho_\theta(s_t) + \epsilon_t, \tag{118}$$

where $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$.

Reward:

$$R(s_t, a_t) = -(s_t - s_{\text{target}})^2 - \lambda a_t^2. \tag{119}$$

Parameters:

- $\alpha \in 0.1, 0.5, 1.0$
- $\sigma = 0.05$
- $s_{\text{target}} = 0$

D.1.2 Tier 2: Multi-Agent Resource System

$$s_{t+1} = s_t - \sum_{i=1}^{N} a_t^i + \beta s_t(1 - s_t). \tag{120}$$

Reward:

$$R_i = a_t^i - \gamma(s_t - s_{\text{safe}})^2. \tag{121}$$

Parameters:

- $N = 5$, $\beta = 0.5$
- $a_{\max} = 0.2$
- $s_{\text{safe}} = 0.5$

D.1.3 Tier 3: Partial Observability

$$s_{t+1} = T(s_t, a_t) + \eta_t, \;\; o_t \qquad\qquad = s_t + \delta_t, \tag{122}$$

with $\eta_t \sim \mathcal{N}(0, \sigma_s^2)$, $\delta_t \sim \mathcal{N}(0, \sigma_o^2)$.
Oracle input is o_t.
Parameters:

- $\sigma_s = 0.1$, $\sigma_o = 0.1$

D.2 Model Architectures

Oracle ρ_θ:

- MLP: [64, 64], ReLU activations
- output: scalar prediction

Policy π_ϕ:

- MLP: [64, 64], ReLU
- outputs: mean $\mu_\phi(r)$ and log-std $\log \sigma_\phi$
- action sampled via Gaussian and clipped to valid range

Value network:

- MLP: [64, 64]

D.3 Training Hyperparameters

- optimizer: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
- learning rate (policy): 3×10^{-4}
- learning rate (oracle): 1×10^{-4}

- discount: $\gamma = 0.99$
- batch size: 2048
- horizon: 200
- iterations: 500
- gradient clipping: $|g|_2 \leq 0.5$

Reflexive term: [$\lambda_{\text{reflexive}} = 0.5$]
Information constraint: [$\beta = 0.01$]

D.4 Training Procedure

Each iteration:

1. Collect trajectories using current policy
2. Compute returns and advantages
3. Estimate gradients (EGP / baseline)
4. Update ϕ and θ

D.5 Evaluation Protocol

- evaluation over 50 episodes (no exploration noise)
- report mean $\pm$ std over seeds
- stability computed per timestep and averaged

D.6 Implementation Details

- framework: PyTorch (v2.x)
- device: single GPU (or CPU)
- state normalization: running mean/variance
- action scaling: tanh-squashed outputs

D.7 Random Seeds

Experiments use fixed seeds: [0, 1, 2, 3, 4]

D.8 Observations

Prediction strength (α):

- small α: near-classical dynamics
- large α: oscillatory regimes

Noise:

- destabilizes classical RL
- mitigated by information constraints

D.9 Reproducibility

All experiments are reproducible given:

- fixed seeds and hyperparameters,
- deterministic environment implementations,
- identical network architectures.

We note that minor deviations may arise due to hardware-level nondeterminism in parallel computation.

E Environment Details

This appendix provides a precise specification of the experimental environments, including dynamics, initialization, constraints, and edge-case handling.

E.1 General Formulation

All environments follow:

$$s_{t+1} = T(s_t, a_t, r_t, \xi_t), \tag{123}$$

where $r_t = \rho_\theta(s_t)$ and ξ_t denotes stochastic noise.

All variables are clipped to predefined ranges after transition updates to ensure numerical stability.

E.2 Tier 1: Controlled Reflexive System

E.2.1 State and Initialization

- $s_t \in [-2, 2]$
- $s_0 \sim \mathcal{U}(-1, 1)$

E.2.2 Dynamics

$$s_{t+1} = \text{clip}\left(s_t + a_t - \alpha r_t + \epsilon_t,\ -2, 2\right), \tag{124}$$

where $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$.

E.2.3 Oracle

$$r_t = \rho_\theta(s_t) \tag{125}$$

During experiments, ρ_θ is implemented as a neural network. For controlled ablations, a linear oracle with additive noise may be used:

$$r_t = ws_t + b + \zeta_t, \quad \zeta_t \sim \mathcal{N}(0, \sigma_r^2). \tag{126}$$

E.2.4 Actions

- $a_t \in [-1, 1]$
- actions are clipped after sampling

E.2.5 Boundary Handling

- additional penalty applied when $|s_t| > 1.8$
- no hard reset is used; trajectories continue under clipping

E.3 Tier 2: Multi-Agent Resource System

E.3.1 State

- $s_t \in [0, 1]$
- $s_0 = 0.5$

E.3.2 Dynamics

$$s_{t+1} = \text{clip}\left(s_t - \sum_{i=1}^{N} a_t^i + \beta s_t(1 - s_t),\ 0, 1\right) \tag{127}$$

E.3.3 Oracle

$$r_t = \rho_\theta(s_t) \tag{128}$$

In bounded settings, a sigmoid transformation is applied:

$$r_t = \sigma(\rho_\theta(s_t)). \tag{129}$$

E.3.4 Agents

- $a_t^i \in [0, a_{\max}]$
- agents act independently conditioned on r_t

E.3.5 Termination

- if $s_t < 0.05$, episode terminates (resource collapse)

E.4 Tier 3: Partial Observability

$$s_{t+1} = T(s_t, a_t) + \eta_t \quad o_t \qquad\qquad = s_t + \delta_t \tag{130}$$

- $\eta_t \sim \mathcal{N}(0, \sigma_s^2)$
- $\delta_t \sim \mathcal{N}(0, \sigma_o^2)$

Oracle input:

$$r_t = \rho_\theta(o_t) \tag{131}$$

E.5 Temporal Structure

- horizon: 200 steps
- discount: $\gamma = 0.99$
- early termination enabled

E.6 Normalization

- states normalized using running mean and variance
- predictions normalized before policy input
- rewards linearly scaled to $[-1, 1]$

E.7 Randomness

- fixed seeds per run
- separate RNG streams for environment and model
- noise sampled independently at each timestep

E.8 Implementation

- vectorized environments (16 parallel instances)
- synchronous rollout collection
- clipping applied after full transition update

E.9 Summary

The environments are designed to expose prediction–action feedback explicitly while maintaining well-defined dynamics under stochasticity, clipping, and boundary conditions. All sources of randomness and numerical handling are specified to ensure reproducibility.

F Metric Derivations

This appendix provides theoretical justification for the evaluation metrics introduced in Section 10. These metrics are designed to capture stability, sensitivity, and information efficiency in reflexive systems.

F.1 Reflexive Stability Error

$$\mathcal{E}_{\text{stab}}(s) = |s - \Phi(s)|. \tag{132}$$

F.1.1 Fixed-Point Residual

Proposition 11. *If $\mathcal{E}_{stab}(s) = 0$, then s is a fixed point of Φ.*

Proof. Immediate: [|s - $\Phi(s)$| = 0 $\implies$ $s = \Phi(s)$.] □

F.1.2 Relation to Distance from Equilibrium

Proposition 12. *Assume Φ is a contraction with constant $L < 1$ and admits a fixed point s^*. Then:*

$$|s - s^*| \leq \frac{1}{1-L}\mathcal{E}_{stab}(s). \tag{133}$$

Interpretation. The stability error controls the distance to equilibrium up to a constant factor.

F.2 Prediction Influence Score

$$\mathcal{I} * \text{pred} = \mathbb{E}\left[|s * t + 1^{(r)} - s_{t+1}^{(0)}|\right]. \tag{134}$$

F.2.1 Operator View

$$\mathcal{I}_{\text{pred}} = \mathbb{E}\left[|\Phi(s) - \Phi_0(s)|\right]. \tag{135}$$

Proposition 13. *If Φ is Lipschitz in the predictive input r with constant L_r, then:*

$$\mathcal{I}_{pred} \leq L_r, \mathbb{E}[|r(s)|]. \tag{136}$$

Interpretation. Prediction influence is controlled by both signal magnitude and system sensitivity.

F.3 Strategic Sensitivity

$$\mathcal{S}_{\text{agent}} = \mathbb{E}\left[|\pi(a \mid r_1) - \pi(a \mid r_2)|\right]. \tag{137}$$

Proposition 14. *If π is Lipschitz in r with constant L_π, then:*

$$\mathcal{S} * agent \leq L * \pi|r_1 - r_2|. \tag{138}$$

Insight. This metric quantifies sensitivity of policies to predictive perturbations.

F.4 Information Efficiency

$$\mathcal{E}_{\text{info}} = \frac{\text{Performance}}{I(S;R)}. \tag{139}$$

Interpretation. The mutual information $I(S;R)$ measures how much predictive signal depends on the state.

Proposition 15. *For fixed performance, reducing $I(S;R)$ improves representational efficiency.*

Connection. This aligns with the information bottleneck principle [17, 4].

F.5 Trajectory Variance

$$\mathcal{V} = \mathbb{E}\left[|s_t - \mathbb{E}[s_t]|^2\right]. \tag{140}$$

F.5.1 Stability Relation

Proposition 16. *If Φ is contractive with constant $L < 1$ and noise is bounded, then:*

$$\text{Var}(s_{t+1}) \leq L^2\text{Var}(s_t) + \sigma^2, \tag{141}$$

where σ^2 is noise variance.

Interpretation. In the absence of noise, variance decays exponentially. With noise, it remains bounded.

F.6 Metric Interactions

The metrics capture related but distinct properties:

- High $\mathcal{I} * \text{pred}$ can increase $\mathcal{V}$.
- Low $\mathcal{E} * \text{stab}$ typically corresponds to stable dynamics.
- Increasing $I(S; R)$ may improve performance but reduce stability.

Insight. Reflexive learning requires balancing competing objectives.

F.7 Limitations of Reward-Based Metrics

Standard RL evaluates:

$$J = \mathbb{E}\left[\sum_t \gamma^t R(s_t, a_t)\right]. \tag{142}$$

This does not capture:

- dynamical stability,
- sensitivity to prediction,
- feedback-induced oscillations.

Conclusion. Additional metrics are necessary to characterize reflexive systems.

F.8 Summary

The proposed metrics are grounded in operator theory, dynamical systems, and information theory. They quantify stability, sensitivity, and efficiency—key properties that are not captured by standard reinforcement learning objectives.

G Stability Analysis

This appendix analyzes stability, oscillation, and sensitivity in reflexive learning systems using linearization and operator decomposition.

G.1 Local Linearization

Let s^* be a fixed point:

$$s^* = \Phi(s^*). \tag{143}$$

Define perturbation:

$$e_t = s_t - s^*. \tag{144}$$

Assuming Φ is continuously differentiable, a first-order expansion yields:

$$e_{t+1} = D\Phi(s^*)e_t + o(|e_t|). \tag{145}$$

Proposition 17. *The fixed point s^* is locally asymptotically stable if:*

$$\rho(D\Phi(s^*)) < 1. \tag{146}$$

G.2 Jacobian Decomposition

Let:

$$\Phi(s) = T\big(s, \pi_\phi(\rho_\theta(s))\big). \tag{147}$$

Assuming differentiability:

$$D\Phi(s) = \frac{\partial T}{\partial s}$$

* $\partial T \frac{}{\partial a \cdot \frac{\partial \pi_\phi}{\partial r} \cdot \frac{\partial \rho_\theta}{\partial s}}.(148)$

Define:

$$J_s = \frac{\partial T}{\partial s},\ J_a \qquad = \frac{\partial T}{\partial a},\ J_\pi = \frac{\partial \pi_\phi}{\partial r},\ J_\rho \qquad = \frac{\partial \rho_\theta}{\partial s}.$$

Then:

$$D\Phi = J_s + J_a J_\pi J_\rho. \tag{149}$$

Interpretation. The second term represents amplification through predictive feedback.

G.3 Instability Condition

Proposition 18. *If*

$$\rho(J_s + J_a J_\pi J_\rho) > 1, \tag{150}$$

then s^* is locally unstable.

Interpretation. Instability arises when feedback amplification dominates intrinsic damping.

G.4 Scalar Case

Consider:

$$\Phi(s) = s + f(s) - \alpha\rho_\theta(s). \tag{151}$$

Then:

$$\Phi'(s^*) = 1 + f'(s^*) - \alpha\rho'_\theta(s^*). \tag{152}$$

Proposition 19. *The fixed point is locally stable if:*

$$|1 + f'(s^*) - \alpha\rho'_\theta(s^*)| < 1. \tag{153}$$

Insight. Increasing prediction sensitivity (α or ρ'_θ) can destabilize the system.

G.5 Oscillatory Behavior

Proposition 20. *If $D\Phi(s^*)$ has complex eigenvalues $\lambda = re^{i\theta}$ with $r \approx 1$ and $\theta \neq 0$, then trajectories exhibit oscillatory behavior near s^*.*

Interpretation. Feedback introduces phase effects that can produce cyclic dynamics.

G.6 Reflexive Gain

Define:

$$G := |J_a J_\pi J_\rho|. \tag{154}$$

Observation. As G increases, the contribution of feedback dominates the Jacobian.

Remark. While precise bifurcation thresholds depend on system-specific structure, increasing G can transition the system from stable to oscillatory or unstable regimes.

G.7 Effect of Noise

Consider:

$$s_{t+1} = \Phi(s_t) + \epsilon_t, \tag{155}$$

with $\mathbb{E}[\epsilon_t] = 0$ and bounded variance.

Proposition 21. *If Φ is contractive, then $\mathrm{Var}(s_t)$ remains bounded. If Φ is expansive, variance can grow over time.*

Interpretation. Instability amplifies stochastic disturbances.

G.8 Multi-Agent Extension

For N agents:

$$D\Phi = J_s + \sum_{i=1}^{N} J_a^{(i)} J_\pi^{(i)} J_\rho. \tag{156}$$

Remark. Under additive interaction assumptions in T, the feedback term grows with N, increasing sensitivity and reducing stability margins.

G.9 Stabilization Insights

Stability can be improved by reducing amplification terms:

- lower oracle sensitivity (J_ρ),
- smoother policies (J_π),
- reduced action impact (J_a),
- regularization or damping.

G.10 Summary

Reflexive systems exhibit structural instability driven by feedback between prediction and dynamics. Stability depends on controlling amplification in the Jacobian, which is governed by the interaction between oracle, policy, and environment dynamics.